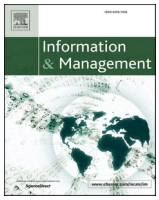
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Leveraging the metaverse ecosystem: How institutional factors, adoption of metaverse-related technologies, and absorptive capacity drive performance in high-tech small and medium-sized enterprises

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ABSTRACT

This study aims to investigate the adoption of Metaverse-related technologies by high-tech small and medium-sized enterprises (SMEs) and its impact on international marketing agility and firm performance. A two-wave longitudinal design was applied using structural equation modeling to assess data collected from high-tech SME executives (n = 226). This study reveals that various institutional aspects positively influence high-tech SMEs' intention to adopt Metaverse-related technologies, leading to its adoption. Adoption improves marketing agility and performance, with absorptive capacity playing a crucial mediating role. Based on the results of empirical investigation, this study provides significant theoretical and practical contributions.

1. Introduction

Modern technology has transformed every aspect of life. The idea of a Metaverse world is fascinating; virtual meetings and interactions using virtual reality (VR) headsets—including via VR games to VR workspaces—are gaining popularity [1,2]. The Metaverse is not a new concept; it is considered a linear process in numerous aspects. Collaborative virtual environments often define virtual worlds as "computer-generated, multi-user, three-dimensional (3D) interfaces in which users can experience the presence of other actors" [3]. Online multiplayer worlds, such as "The Sims" or "Second Life," have existed for nearly two decades, and players spend an average of 20 h a week in these worlds [4]. The existing services and business models may survive in the Metaverse, but new interactions may lead to further operations and business patterns. Not everything in the Metaverse applies to all industries. The Metaverse's effectiveness stems from its participatory nature and personalized user acceptance [5]. Consequently, a gap exists in seizing the opportunity to investigate the adoption of Metaverse-related technologies (AMT), particularly among high-tech small and medium-sized enterprises (SMEs).

The Metaverse is not a single technology but a blend of related technologies (e.g., augmented reality [AR], VR, IoT, extended reality (XR), blockchain, artificial intelligence (AI), machine learning (ML), nonfungible tokens (NFTs), digital twins, and other networking technologies) that operate jointly to develop and empower a cohesive and immersive virtual world [4,6]. Invariably, innovation occurs everywhere, which is thrilling because we do not know what lies next. It is crucial for businesses to strategically assess these technologies and consider how well they fit with their target markets and business objectives. Beyond this hype, Metaverse-related technologies have the potential to radically transform how we operate businesses and communicate worldwide [7]. One study predicts that by 2026, 30 % of global enterprises will offer Metaverse-ready products and services [8]. Furthermore, it is anticipated that by 2026, 25 % of the global population will spend at least 1 hour every day in the Metaverse for activities including work, education, entertainment, commerce, and socialization [9]. According to estimates based on the potential impact of this new digital realm, the Metaverse could contribute an additional US$5 trillion to the global economy by 2030 [10]. In addition, Statista predicts that the Metaverse market will reach $74 billion by 2024 and is expected to reach 263.3 million users by the end of 2030 [11]. The economic impact of Metaverse depends on the speed of technology development, the adoption of related technologies (such as AR, VR, IoT, MR, blockchain, AI, ML, NFTs, digital twins, and other networking technologies), and the ability to address interoperability challenges. A recent study investigating the adoption by organizations noted the important roles of digital agility and the abundance of innovation theories in facilitating the use of Metaverse by businesses [12]. On the other hand, Kumar et al. [13] pointed to psychological, security, and performance issues as major

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List of Abbreviations

Abbreviations	*Definition*
AR	Augmented Reality
VR	Virtual Reality
XR	Extended Reality
PSEB	Pakistan Software Export Board
IMA	International Marketing Agility
AMT	Adoption of Metaverse-related technologies
IAMT	Intention to adopt Metaverse-related technologies
AbC	Absorptive Capacity

barriers to adopting the Metaverse and offered a strategy to circumvent them. Meanwhile, in terms of user experience, Kumari et al. [14] found that while technical difficulties hinder user engagement, visually pleasing environments, watch ad-to-earn features, and social interactions positively impact user intention toward the Metaverse platform. Therefore, considering the rapid growth and importance, SMEs should consider utilizing Metaverse-related technologies, which may enable them to reach new clients, extend their business horizons, and enhance their performance. Our research focuses on how high-tech SMEs strategically adopt these foundational technologies to meet future needs and reap marketing and performance benefits from the imminent Metaverse environment.

Implementing these technologies can position SMEs for future success and help them stay ahead of the curve in a quickly changing digital environment [15]. In an immersive Metaverse ecosystem, technologies like digital twins enable users to manipulate complex systems in a controlled virtual space, increasing efficiency and making informed decisions through real-time monitoring and predictive maintenance. This also facilitates the global collaboration of virtual models, revolutionizing design, training, and operational strategies in sectors such as manufacturing, healthcare, entertainment, and urban planning [16,17]. Meanwhile, effective implementation of Metaverse can support an enterprise's various advertising, branding, and marketing activities. For example, businesses can benefit from immersive marketing campaigns to allow customers to interact with a product, brand, or service virtually [18]. Adopting Metaverse also helps provide a space to socialize and enhance visual appeal, which helps improve SMEs' financial and market performance [19]. High-tech SMEs, in particular, will benefit greatly from the virtual world's innovative potential. However, the pace of technological development poses a significant barrier, as it often requires the assets and skills of large-scale firms. Understanding the proper adoption and utilization of Metaverse-related technologies by high-tech SMEs is critical to their continued prosperity and competitive advantage. Owing to their infancy, enabling technologies still may need to be widely adopted. Despite the hype surrounding Metaverse, there is a notable lack of research on its adoption in the SME sector, especially among high-tech SMEs. This study has the potential to fill this gap by aiding high-tech SMEs in making knowledgeable decisions on adopting Metaverse-related technologies in their businesses and accelerating their performance.

This study adopts an institutional isomorphism perspective—assuming firm autonomy—to explain SMEs' adoption of technology, which differs from the prevailing assumptions underlying the studies on innovation diffusion. The purpose, characteristics, and pitfalls of the three institutional isomorphic processes—along with their impact on internal business processes and the sharing of technological resources between firms—have been discussed previously [20–22]. Technology adoption is often considered when establishing high-tech SMEs, particularly in developing nations [23]. High-tech SMEs may have considerable competitive advantages if they early adopt Metaverse-related technologies. This will enable them to stand out and innovate in an increasingly digital marketplace. Owing to external pressures and legitimacy issues, it is argued that institutional factors are a prerequisite for high-tech SMEs' intention to adopt Metaverse-related technologies (IAMT). Unlike economic motivations, institutional factors include regulatory requirements and industry standards that are critical to global business. As Metaverse-related technologies spread, institutional pressure is created for firms to adopt them to remain legitimate and competitive. Although several theories are widely used to examine technology adoption in other contexts [24,25], as well as in context to AMT across various fields [26–31], the high-tech SME literature has largely ignored institutional forces. In the context of the AMT in high-tech SMEs, this study fills a gap in the literature by extending the concept of institutional isomorphism from the level of personal decision making inside a firm to the level of interorganizational processes.

Moreover, growing evidence suggests that the perception of the presence or absence of several users considerably impacts decisions regarding new technology adoption [32,33]. This enables us to contribute to the established concepts of the emerging pattern of perceived critical mass (PCM), which may be integrated with institutional theory. Contributions to this theory focus on PCM, which is not yet been studied extensively in the information systems literature [34]. Furthermore, adopting technical breakthroughs may result in major changes in an organization's internal and external operations [35]. Institutional factors considered important determinants of technology adoption include organizational readiness (OR), which is consistent with other institutional isomorphisms. Therefore, SMEs need to know whether they are ready to adopt innovative technologies related to Metaverse. Similarly, with the rise of uncertainty and competitiveness in global marketing, agility has emerged as a potential alternative to conventional planning and execution processes [36]. Furthermore, although the general marketing literature assumes that agility improves performance [37,38], empirical support from the perspective of international marketing needs improvement. One study defined the key dynamic capabilities that are tacitly responsible for high-tech SMEs' international agility [39]. Similarly, the need to acquire and manage knowledge to obtain a competitive edge has been emphasized in the literature [40,41]. Therefore, absorptive capacity (AbC) enables businesses to identify, collect, analyze, understand, and creatively use external information to help management create a firm's competitive edge [23,40,42]. This motivates us to identify the role of international marketing agility (IMA) and AbC in overall firm performance (FP) in the context of high-tech SMEs, which has yet to be studied.

Considering the potential of Metaverse-related technologies to revolutionize marketing operations and the absence of systematic inquiry and preliminary empirical analysis of Metaverse research from an IS viewpoint in SMEs, this study addresses the following research questions:

RQ1. What key factors influence high-tech SMEs' IAMT?
RQ2. What is the impact of AMT on IMA and FP in high-tech SMEs?
RQ3. What is the effect of SMEs' AbC to leverage AMT on their IMA and FP?

This study conceptualizes and empirically validates a research model that identifies the key factors generating adoption intention leading to AMT and its impact on the IMA and FP of high-tech SMEs in Pakistan. The remainder of the paper is divided into the following sections: Related work and theoretical background are presented in Section 2. The research model and hypotheses are elucidated in Section 3. The research methodology is described in Section 4. The study's results are presented in Section 5. The discussion, theoretical and practical implications, limitations, and prospects are outlined in Section 6. Conclusions are provided in Section 7.

Table 1
Key technologies related to the Metaverse.

Technologies	Description	Examples	Practical use for SMEs
Virtual reality (VR)	Immersive digital environments are experienced through VR headsets.	Oculus Rift, HTC Vive, PlayStation VR	A virtual tour to provide potential customers with an immersive product demonstration.
Augmented reality (AR)	Overlaying digital information in the real world through devices like smartphones or AR glasses.	Pokémon GO, Microsoft HoloLens, Google Glass	Interactive marketing materials, enhanced navigation systems, and virtual try-on solutions for retail (e.g., eyewear, apparel).
Mixed reality (MR)	Blending both physical and digital worlds, where digital objects interact with the real world in real-time.	Microsoft HoloLens 2, Magic Leap2	Virtual collaboration in the workplace, interactive product demonstrations in retail.
Extended reality (XR)	An umbrella term that encompasses VR, AR, and MR, representing all immersive technologies.	Combination of VR, AR, and MR devices	Comprehensive training environment with various XR elements.
Blockchain	Decentralized technology ensures security and transparency in digital transactions and assets.	Cryptocurrencies (Bitcoin, Ethereum)	Secure transactions in virtual marketplaces, transparent supply chain management, and decentralized finance.
Nonfungible tokens (NFTs)	Unique digital assets verified using blockchain technology, representing ownership of virtual items.	Digital art, virtual real estate, in-game items	Authenticating ownership of digital assets and trading virtual items.
Artificial intelligence (AI)	Technologies that enable machines to simulate human intelligence and problem-solving capabilities enhance the user experience with intelligent systems that can learn and make decisions.	Virtual assistants (Siri, Alexa)	Personalized virtual shopping assistants, AI-driven characters in virtual worlds, personalized marketing campaigns, inventory management, etc.
Machine learning (ML)	A subset of AI that focuses on developing systems and algorithms that learn and improve from data enables AI to imitate how humans learn and gradually improve its performance accuracy.	Recommendation systems (product recommendation), statistical arbitrage, predictive analytics	Tailoring user experiences in virtual worlds, improving AI characters, sales forecasting, and customer behavior analysis.
Internet of Things (IoT)	IoT incorporates electronics, sensors, communications, and computer science engineering, which assists in connecting physical devices to the Internet to collect and exchange data.	Smart home devices, wearable technology, connected equipment	Integrate physical and virtual environments to collect data for virtual simulations to monitor and optimize energy use, remote asset management, and improve workplace safety.
3D reconstruction	Creating 3D models of real-world objects and environments for virtual representation.	3D scanning technologies, photogrammetry software.	Creating realistic virtual environments and digitizing real-world artifacts for products and services for e-commerce.
Spatial computing	Interact with digital and physical space using spatial awareness technology and human-computer interaction within 3D environments.	VR/AR spatial mapping, gesture recognition systems, Vision Pro, and Meta Quest Pro	Enhance customer engagement by creating realistic virtual environments and real-world artifacts of digitized products and services.
Digital twins	A digital model provides the virtual representations of physical objects or systems used to simulate, predict, and optimize performance.	Digital models of factories or cities, Facilities management	Simulating production processes, monitoring equipment health, optimizing logistics and supply chain operations.

2. Literature review and theoretical background

2.1. Metaverse technology

Metaverses are virtual worlds that provide additional dimensions and realms for economic, social, and entertainment acts in our real world [43]. Neal Stephenson represented how the VR-based Internet would evolve in the future [44]. New definitions and concepts develop as the Metaverse's breadth broadens [45]. A 3D virtual environment wherein virtual characters engage in political, economic, social, educational, and cultural activities is called a Metaverse [46]. The Metaverse is a 3D immersive virtual environment that allows users to engage in virtual computational social and economic activities. It refers to the process of duplicating the digital version of a person's actual physical world and then adding it to the analog data [47]. In other words, a mirror world mimics a real-world structure or item's appearance while retaining its qualities and functionalities [48].

The Metaverse differs from AR, VR, and XR. AR uses digital overlays of digital content to improve real-world experiences, while VR provides a fully analog and extremely immersive experience that requires specialized headsets to replace the real world. In addition, XR acts as an umbrella, allowing for transition or mixing between AR and VR to provide multiple immersive experiences. The Metaverse represents a broader concept of collective virtual shared spaces that integrate VR, AR, and other technologies into a persistent, interconnected digital universe that allows people to interact and participate in various activities [49]. The literature also differentiates this concept. First, most VR-related research is conducted in the realm of physical methods and visualization; however, the Metaverse possesses strong qualities as a service with long-term substance and contextual significance. Second, AR, XR, and VR are used only sometimes in the Metaverse; however, a Metaverse application can exist even if the platform does not permit VR or AR interaction. Finally, the Metaverse features an extensible environment that can handle several individuals, essential for enhancing its social significance [48]. Irrespective of how futuristic this seems, over the years this technology's potential to divide the virtual and real physical worlds has become evident [45]. Unsurprisingly, virtual worlds, such as "Second Life," have been used to sell and promote both virtual and real-world items, with supply networks connecting several platforms. However, Metaverse transactions expand economic activities' scope, scale, and context [50]. Similarly, the Metaverse must extend the "traditional" taxonomy, integrating a different dimension to how the Internet creates environments for electronic transactions [51]. These environments describe and investigate human behavior in industrial settings to identify how it varies from the actual world [23]. Table 1 lists and describes the potential industrial uses of Metaverse-related technologies.

These technologies share basic characteristics, even as they have broad applications in the Metaverse ecosystem that include virtual environment interaction, interoperability, digital asset management, interaction design, and immersive user experiences. Although functions of Metaverse-related technologies may vary, the broader context of technological infrastructure and enterprise strategy often overlap. This study posits that high-tech SMEs encounter similar drivers and barriers when adopting these cutting-edge technologies, regardless of whether they are adopting specific technologies related to the Metaverse. Therefore, the variables that affect the adoption of one Metaverse-related technology also may affect other technologies.

Furthermore, the age of digital dependence has transformed consumer behavior. Studies have shown that the COVID-19 pandemic has increased the number of individuals buying, working, gaming, and learning online [52,53]. Marketers can use the Metaverse to learn about consumer preferences and build immersive, interactive advertising experiences that delight consumers [54]. The Metaverse claims to offer

physical and digital solutions that blur the barriers between actual reality and VR and alter our perceptions of the environment [55]. According to Dwivedi et al. (2022), who examined executive's perspectives on the Metaverse's foreseeable societal impacts, this technology will open vast new avenues for businesses to advertise their products and services, provide students with more in-depth learning opportunities, introduce new avenues for tourists to visit remotely from anywhere in the world, and usher in a new era of travel [7]. Overall, the Metaverse has significant commercial potential. The optimal meta-marketing strategies can help organizations gather valuable insights [56]. SMEs should adopt Metaverse-related technologies to improve consumer interactions and promote international marketing. They must consider constructing a sophisticated digital factory or industrial space in the spectrum and exploring how exoskeletons interact with the physical world [4]. Moreover, the Metaverse can extend consumer access to marketplaces in emerging and frontier countries.

Although companies have recently reduced their direct spending on Metaverse platforms and virtual worlds, organizations like Apple and Nvidia still prioritize investments in Metaverse technologies [57]. This nuanced perspective illuminates a complex view of technological innovation and highlights the multifaceted nature of enterprises' adoption of virtual environments. However, some companies may need more support to embrace these virtual environments fully. Practical application of the Metaverse takes time to penetrate the developing world; however, firms are actively exploring and investing in technologies that support the Metaverse ecosystem. Owing to the data availability and its impact on economic development, prior research on innovation efficiency at the firm level focuses mainly on large-scale listed firms. Scholars have been expanding their research to include emerging technologies in the Metaverse to enhance their understanding of the dynamics of innovation [58]. Therefore, we propose that high-tech SMEs can position themselves for success in the future and stay ahead of the curve in a rapidly changing digital environment by implementing these technologies in a virtual setting.

Prior Metaverse research has focused on virtual world configurations (e.g., games); however, more recently, the Metaverse has been described as a content-focused platform for exchanging interests and facilitating social engagement [59,60]. Using the technology acceptance model, Akour et al. [5] demonstrated how the Metaverse-based education system's perceived usefulness influences university students' intention to adopt it. Similarly, the key role of social and psychological factors in Metaverse adoption in various situations also has been discussed in previous studies [61–63]. Given this, Abumalloh et al. [64] examined factors that influence adoption of the Metaverse in retail based on the resource-based view theory [64]. A prior study examined Metaverse transportation systems, such as AI-based autonomous driving, Metaverse public transit, AI-managed traffic, and rental or loaned cars [65]. Jaung (2022) used a discrete choice experiment with university students to examine their interest in digital forest tourism in the Metaverse [60]. Similarly, studies investigating the transformational potential of Metaverse technology in the tourism industry [66,67] and the hospitality industry [68] have provided new insights and identified basic areas for improvement. The findings of a recent study clarify important concerns for organizations trying to establish strategic positions in diverse demographic groups while examining the behavioral intentions of Generation Z's adoption of the Metaverse [69]. Moreover, to provide insight into customized experiences, Mirza et al. (2024) categorized potential users into skeptical, unaware, and optimist [70]. Similarly, Natarajan et al. (2024) suggested the importance of improving easy navigability, escapism, social interaction, and telepresence as key factors in adoption of the Metaverse [71]. These results provide practical insights into improving user engagement and experience, which may encourage wider AMT. Despite the growing importance of the Metaverse and related technologies, to our knowledge, further evidence for SMEs in needed, in the context of establishing a relationship with AMT that enhances FP in emerging economies. Few studies to date have suggested that AMT can boost customer involvement, improve employee productivity, and reduce costs, such as fuel and rent costs [54,59,72]. Table 2 provides an overview of Metaverse technology adoption across various fields.

Key topics of these studies include barriers to adoption, the impact of psychological and social variables, factors influencing Metaverse adoption, and the Metaverse's impact on supply chains, gaming, retail, hospitality, and education. However, from an institutional theory perspective, research on Metaverse-related technology adoption highlights research gaps in how normative pressures, regulatory frameworks, and mimetic behavior influence the organization's adoption decisions. Similarly, little is known about how OR and PCM affect institutionalization of the Metaverse ecosystem. There also is a dearth of research on how AMT helps organizations gain legitimacy and how various institutional factors influence adoption choices in achieving IMA and improving FP. Moreover, the extent of demand for Metaverse-related technologies by high-tech SMEs remains unclear. Clarifying this demand is essential for investigating the influence of institutional and other supporting factors on the IAMT and its potential impact on enhancing businesses' dynamic capabilities and performance. Filling these gaps allows for a more complete understanding of the institutional factors that support or hinder AMT in high-tech SMEs.

2.2. Dynamic capability theory

According to dynamic capability theory, an enterprise's ability to continuously improve and adjust its resources and capabilities is critical for long-term success [73]. In strategic management, AbC is closely related to dynamic capability theory. Zahra et al. (2002) conceptualized AbC as a dynamic capability that enhances the opportunities for a business to gain and maintain competitive advantages [74]. The literature has considered AbC to be a key component of the dynamic process that enables enterprises to identify and exploit new growth and innovation opportunities, which enhances their performance [75,76]. Therefore, this study emphasizes AbC as a fundamental dynamic capability for high-tech SMEs seeking to adopt Metaverse-related technologies and successfully utilize them to achieve a competitive advantage. The AbC was first proposed by Cohen et al. [77], who defined it as "the ability of a firm to recognize the value of new external information and assimilate and apply it to commercial ends." It is a relative construct that depends on the context of the social relations in which knowledge is embedded [78]. The AbC theory proposes that knowledge gained from previous experience helps identify, select, and implement practices within a business to create a viable competitive edge [79]. A firm's information absorption capacity is measured by its effectiveness in transformational, developmental, and exploratory learning processes [40]. Management may support or allow the use of new information technologies, indicating that senior executives can implement new technology [80] and innovation strategies [81], such as AMT.

Zahra et al. (2002) described two categories and four aspects of AbC [74]. These two categories are potential AbC (knowledge acquisition and assimilation) and realized AbC (transformation and exploitation of assimilated knowledge). Similarly, according to Hötte [82], AbC is revealed by the existence of four elements: "knowledge acquisition," "knowledge assimilation," and the "transformation" and "exploitation" of assimilated knowledge. Studies have proven that these dimensions of AbC play different but complementary roles in organizational performance [83,84]. In conclusion, AbC is vital for the acquisition, assimilation, transformation, and exploitation of knowledge. The studies prove that firms with strong AbC exhibit strategic flexibility and can respond strategically to business risks and opportunities [23,85].

In the context of the current study context, AbC is seen as a potential organizing capacity that manifests through various knowledge-related processes rather than the capacity formed by aggregating these processes. Therefore, we measured AbC (as a higher-order reflective construct) based on these dimensions as a higher-order reflective scale.

There is a dearth of research on the connections among AbC, IMA, and FP. Considering the significance of AbC, as indicated by previous studies, this study finds the AbC to be the most critical aspect in integrating technology adoption, IMA, and FP.

2.3. Institutional isomorphism

Institutional theory suggests that technology adoption decisions are based on rational considerations and the external environment's social and normative pressures. It comprehends the organization as a manifestation of social ideals and a distinctive social system directed by internal and external institutional judgments [20,86]. DiMaggio and Powell (1983) defined institutional isomorphism as "the major factors that organizations must consider are other organizations" [87]. They identified three institutional isomorphisms influencing technology adoption: "mimetic," "coercive," and "normative." Mimetic isomorphism (MI) may lead organizations in the same institutional environment to adopt similar technologies or management methods to avoid risky decisions and conform to conventions, practices, and values. This can homogenize organizational behavior and decision making through the adoption of identical technologies. Additionally, isomorphism may arise from regulations that entail adopting a certain level of technology or providing legitimacy. Coercive isomorphism (CI) refers to a possible preferential treatment for organizations that respond to technology adoption. Normative isomorphism (NI) arises from the need to legitimize organizations based on norms established by trade associations, educational institutions, or accreditation procedures [87]. In conclusion, the institutional theory proposes that businesses within the same institutional environment are more likely to adopt similar technologies owing to shared social and normative incentives to conform to long-standing beliefs and practices.

Institutional isomorphism represents an organization's structure shift as it attempts to adjust rationally to the organization and its environmental uncertainties and constraints [22]. An organization's external environment comprises a range of actors—including suppliers, consumers, governing bodies, and partner organizations—in a standard institutional camp [88]. Existing technology acceptance models based on innovation diffusion theory assume a vast group of influencing factors, define them as perceived rather than existing, and fail to explain which individual players generate them [20]. In this vein, institutional theory defines the drivers that motivate a firm toward innovation and technology adoption, and various scholars have used it in different contexts of technological adoption [89–91]. The high-tech industry addresses these factors in the current research context, wherein technology adoption is based on institutional pressure to lead the market share. Furthermore, the institutional theory posits that individuals conform their innovative behavior to their institutional settings' prevailing norms and values [92]. The institutional theory argues that the need for legitimacy is the primary force driving organizational practices (e.g., using Metaverse-related technologies).

Similarly, institutional isomorphism also may be applied to AMT, whereby firms may adopt this technology to adhere to their institutional environment's norms, practices, and beliefs, to avoid being perceived as deviant or uncoordinated with their peers. Businesses already praise the Metaverse for better promoting their products and services, keeping up with market developments, and investing in creating an immersive digital world [6]. Because institutional theory captures the social and institutional constraints that influence these decisions—pressures that economic theory sometimes ignores—it is particularly useful for understanding the motivations that drive SMEs to develop their intentions, considering that transaction cost economics, resource-based view, and other economic theories have not solved the isomorphic constraints faced by SMEs. These theories attach great importance to cost–benefit assessment and strategic resource allocation. In addition, institutional theory emphasizes the importance of consistency and legitimacy, both of which are critical for SMEs seeking to gain the trust of stakeholders and customers. Considering the current study's perspective, the institutional theory is applied to demonstrate that regulatory pressures cause firms to become more comparable and develop their IAMT. This suggests institutional isomorphism (mimetic, coercive, and normative) influences practices within and across high-tech SMEs. We believe that this approach leads to a deeper understanding of the motivations behind AMT in this context. Therefore, institutional isomorphism is considered this study's supporting theory.

Although institutional theory provides a strong framework for understanding how mimetic, normative, and coercive forces influence organizational behavior, it might not comprehensively account for the internal perspectives that play key roles in adopting new technologies such as the Metaverse. The assessment of the OR of SMEs is necessary because it determines whether they have the infrastructure, resources, and strategic alignment needed to successfully adopt and execute Metaverse-related technologies. In the absence of such internal capacity, even strong institutional pressure might not lead to adoption. Similarly, SMEs' assessment of how well the technology is accepted in the industry is reflected in the PCM, which can greatly impact an SME's adoption intention. This concept is especially important for new technologies, such as those in the Metaverse ecosystem, as their use and value will grow as more firms adopt them. Therefore, this study recognizes that incorporating OR and PCM into the research model is critical for determining IAMT that will not be driven solely by external pressures. Under the podium of institutional theory, this integrated approach provides a more comprehensive understanding of the adoption process. It ensures that the model thoroughly addresses the internal and external forces that initially influence the adoption intentions of Metaverse-related technologies in high-tech SMEs.

3. Hypotheses development

3.1. Impact of institutional isomorphism

When a firm's goals are unclear and its operating environment is unstable, it follows other organizations—especially those closely related to it—in managing an uncertain business environment [22]. As the imitated firm may be a handy source of practice, followers may be unaware of their copycat behavior. Initiators may miss creating alliances with their suppliers and consumers, who may struggle to justify the accompanying expenditures required to achieve technology adoption benefits [20]. Firms operating in comparable surroundings tend to imitate different corporations' moves, primarily based on the numbers eliciting the behavior. This mimicking behavior prevents the guilt of being less responsive or inventive than competitors [93]. A recent study highlighted that MI levels positively correlate with the adoption of agile innovation management by SMEs [94].

Imitation of the Metaverse by other firms among high-tech SMEs can serve as an innovation incentive, enabling firms that still need to adapt it to replicate the behavior of adopting Metaverse-related technologies. After observing the positive effects of Metaverse technologies' on other businesses, we believe adopting them seems beneficial. A newcomer can follow the incumbent's path. As it becomes increasingly widespread, numerous businesses may adopt this technique without much consideration. Firms are driven by their competitors' apparent success and planned actions, which supports the concept that using Metaverse technology is necessary to maintain a competitive advantage. In sum, MI can strongly impact the development of firms' IAMT. Therefore, we propose the following hypothesis:

Hypothesis 1a. MI positively impacts the IAMT.

3.1.1. CI

CI stems from political influence and legitimacy. Classical Institutional Theory describes CI as originating from institutions that impose regulations on businesses [87]. The cultural expectations of other

organizations and societies wherein the affiliation operates exert CI on the association [94]. Firms depending on the dominant enterprise for survival follow the requirements directly imposed by the dominant partner and adopt the technology [95]. While incumbents assist with technology adoption, firms adopting technology are also at risk of revealing their internal processes and trade secrets to incumbents [22]. The study [96] highlights that organizations with a high degree of CI are more likely to adopt agile innovation. Prior literature has proven that CI positively correlates with technology adoption [80,97].

In this study's context, institutions—such as federal and local governments and other potentially resource-advantaged parties—use their influence to actively limit high-tech SMEs by compelling them to engage in internationalization processes by adopting advanced technologies. In the AMT, a firm may be vulnerable to CI. If competitors successfully adopt Metaverse-related technologies, some firms may feel forced to embrace them. If a sector or industry has few firms adopting Metaverse-related technologies, the pressure on the remaining firms to do so may decrease. Therefore, industries may adopt this technology slowly. Therefore, firms under such pressure may be more inclined to show a greater desire to implement metaverse-related technologies to comply with external standards and avoid consequences. Thus, CI may strongly influence a firm's IAMT, which improves industrial performance and has a substantial economic impact. Therefore, we hypothesize as follows:

Hypothesis 1b. CI positively impacts the IAMT.

3.1.2. NI

NI refers to the influence of preexisting standards established by professionals and industry groups [98]. Professional associations, industry standards, and the media are sources of NI [99]. Firms that follow industry standards have knowledgeable and amicable relationships with consulting firms that share the benefits and beliefs of adopting new technologies [100]. Firms conform because they believe that complying with norms established by industry groups and professionals benefits their businesses [94].

No cohesive pressure is involved in transmitting NI; corporations adhere to standards because decision-makers perceive themselves as part of specific institutional communities. This pressure is a consequence of greater professionalization. Business executives agree that adhering to the guidelines established by the relevant professional and industrial associates is appropriate. Moreover, previous studies have investigated NI as a potential indicator of technological adoption in various settings [101,102]. Firms may feel pressure to adopt Metaverse-related technologies to comply with industry norms, follow best practices, or meet the expectations of stakeholders and peers. This societal pressure may greatly influence their decision to incorporate related technologies into their operations, which will increase adoption rates. In this study's context, NI may play a significant role in shaping the IAMT in high-tech SMEs. Therefore, we propose the following hypothesis:

Hypothesis 1c. NI positively impacts the IAMT.

3.1.3. OR

OR is described as an organization's estimation of its readiness to absorb, adapt, and utilize—and eagerness to embrace—new, innovative technologies [103]. The study [104] describes OR as "managers' perception and assessment of the level of awareness, resources, commitment, and governance they believe their organization has to adopt information technology." This standard primarily focuses on technical and financial readiness. Technology readiness refers to an organization's perception of its specialized human resource capabilities and technology-related infrastructure [105]. The term "financial readiness" describes an organization's attitude toward resource availability and smooth transformation [96]. Financial resources are used for ongoing expenditure during cloud computing implementation and use. Meanwhile, organizations with adequate infrastructure, personnel competence, and financial support should boost technology utilization [106].

Changing the behavior of employees and other key stakeholders is difficult, especially in SMEs, if the OR for implementing agile innovation management is insufficient [94]. The present viewpoints are compatible with the idea that "readiness for change" is a prerequisite for effectively implementing complicated changes. When discussing a firm's readiness for change, considering the level of change in the commitment and efficacy among its members is important. Metaverse technologies require resources, skills, and expertise to boost deployment efficiency and adoption. However, organizational passiveness makes it difficult to generate compelling intentions to adopt Metaverse-related technologies. Similarly, if Metaverse-related technologies do not fit the firm's goals or critical stakeholders do not support it, building a positive intention to adopt it becomes challenging. Moreover, compared to traditional technologies, Metaverse-related solutions may require a significant investment in specialized software, hardware, and staff with knowledge of the Metaverse ecosystem. Firms are more likely to get the advantages of adopting meta-related technologies and are more confident of successfully executing them if they are well prepared, including having adequate financial resources, technical know-how, and strategic alignment. Therefore, in this study's context, we can assume that OR is a crucial predictor of IAMT in high-tech SMEs. This leads us to the following hypothesis:

Hypothesis 2a. OR positively impacts the IAMT.

OR for change theory states that higher OR increases the success rate and lowers the risk of failure for innovation adoption [107]. Consequently, readiness models must consider context and be adapted to appropriate areas [108], that is, a specific technology, such as Metaverse. Assumedly, a firm's capacity to accept and successfully implement new technology depends on several interrelated aspects, including cultural and personnel considerations, as well as the degree to which existing technological capabilities align with the desired outcome [109]. Therefore, this will precipitate a more effective adoption of innovative technologies, such as Metaverse-related technologies. Similarly, existing theoretical studies have demonstrated that OR significantly impacts the adoption of novel technologies such as blockchain [35,110] and artificial intelligence [111].

Metaverse-related technologies have not yet achieved significant penetration, and the firm's hesitation to adapt may influence its intention to adopt the technology. Therefore, individuals have long been considering OR for using new technologies, both in industry and academia. Furthermore, if high-tech SMEs are still not ready for such technologies, it may be difficult to adopt them, thus causing delays and failures in their prospects. In contrast, a firm that is prepared for AMT can utilize its benefits faster and better than one that is not. However, OR's precise impact on AMT remains unclear and must be confirmed through empirical investigation. Therefore, we propose the following hypothesis:

Hypothesis 2b. OR positively impacts AMT.

3.1.4. PCM

PCM—initially described as "a small segment of the population that chooses to make big contributions to the collective action" [112]—often provides the necessary conditions for cooperation and coordinated action [34]. This idea was eventually applied to studying invention diffusion, defined as "the point at which enough individuals have adopted an innovation so that the innovation's further rate of adoption becomes self-sustaining" [113]. As the value of interactive communication platforms expands exponentially with the number of users, critical mass theory becomes particularly essential in understanding its acceptance and use [114]. However, PCM has long been deemed one of the most significant and crucial elements in determining communication technology uptake [115]. Moreover, individuals' subjective perceptions of technological innovation's PCM affect their beliefs regarding the

creation and future adoption behaviors. Further, previous research has provided strong evidence for conclusions regarding PCM and its impact on technology adoption [32,116]. Therefore, knowing how many people are using the technology is likely to influence a person's intention to use the technology [32].

In this study's context, if high-tech SME' executives perceive that most of their stakeholders use Metaverse-related technologies in their businesses, they must develop their IAMT. A high-tech firm may be reluctant to adopt Metaverse-related technologies if it does not observe widespread adoption among its peers. They may assume that the Metaverse does not have enough users to interact with or that the technology is still not ready. Thus, firms should raise awareness and educate the market regarding Metaverse-related technologies' advantages and applications to increase the CM and IAMT. Therefore, we posit the following hypothesis:

Hypothesis 3. PCM positively impacts the IAMT.

3.1.5. IAMT

An individual's actual use of technology may often be predicted by their behavioral intention, that is, plans for how they will behave in the future [117]. The intention to change one's behavior is the first step toward successfully implementing any change. As stated by [118], behavioral intention plays a pivotal mediating role between various predictors and the adoption of multiple systems. Prior literature has established that a person is more likely to accept a technology if they intend to use it [119]. The social sciences literature has clarified that the intention to adopt technology directly impacts adoption, and numerous studies have confirmed its significance [30,120]. The important factors influencing adoption include perceived ease of use, perceived usefulness, compatibility with current systems and infrastructure, and perceived danger [119]. However, these are not the only factors to consider when assessing the extent to which technologies are adopted.

We argue that adoption decisions are anticipated to be increasingly influenced by institutional factors when Metaverse-related technologies reach the mainstream and become widely available. Advancements have substantially aided the concept of the Metaverse in cloud computing, wide-area network connectivity, AR, VR, XR, digital currency, and NFT [7]. In a report by [121], a scenario outlining various adoption possibilities is presented. Several perspectives exist, from a correctly structured paradigm to a low-adoption, fragmented market model to full integration of the real and virtual worlds in the Metaverse. In the case of Metaverse-related technologies in high-tech SMEs, the intention to adopt it would include investigating the numerous possible Metaverse platforms and developing a strategy to incorporate the technology into business and vice versa. Consistent with previous studies and the current discussion, our analysis considers the IAMT an active contributor to AMT in high-tech SMEs.

Hypothesis 4. IAMT positively impacts AMT.

3.1.6. AMT

Technology not only assists businesses in becoming more efficient and pragmatic, but also directly affects their capacities, such as AbC [122]. Moreover, organizations can attain their goals if they adequately execute appropriate technology-based strategies [123]. Daniel et al. [124] posited that the link between ICT and AbC is critical for industries to create additional knowledge resources and improve their latent expertise to achieve performance goals. Dwivedi et al. [7] identified various issues stemming from the widespread adoption of the Metaverse and proposed new research directions by integrating several important factors.

The Metaverse potentially increases firms' AbC for new information by providing them with narrative opportunities to achieve their goal in a virtual setting. Furthermore, Metaverse firms may improve their AbC-relevant information by collaboratively sharing best practices and developing new products and services. By offering virtual learning and collaboration, Metaverse-related technologies may improve high-tech SMEs' AbC by helping them locate, grasp, and use new knowledge, practices, and technology. By adopting Metaverse-related technologies, high-tech SMEs can connect with other stakeholders and subject matter experts and gain access to previously inaccessible information and knowledge. Therefore, we presume that AMT may positively impact AbC and formulate the following hypothesis:

Hypothesis 5a. AMT positively impacts AbC.

IMAT reflects an organization's ability to act and respond flexibly in the face of sudden changes in the international market [125]. Significant outcomes have been achieved in terms of how technology adoption supports fluid and reliable access to external information [23]. This may help firms gain marketing agility on a global scale and improve performance. Using technical tools helps promote knowledge assimilation, acquisition, transformation, and exploitation, thereby improving innovation and performance [75]. Nevertheless, integrating innovative technology depends on several elements, such as the resources available to incorporate the technology or infrastructural needs. Firms and stakeholders must overcome these barriers to assess information coherently, efficiently, and effectively, such as to influence the development of their agility and performance [83]. Thoumrungroje and Racela [125] conceptualized the positive relationship between IMAT and new technology adoption among SMEs.

AMT allows businesses to construct realistic and interactive virtual environments for customers, which can improve customer involvement and help firms adapt to the changing markets. Metaverse-related technologies enable global-virtual commerce, showrooms, and events, which reduces travel, helps firms approach new markets, and provides real-time customer behavior data for developing marketing strategies. They also enable firms to personalize virtual experiences for different cultures and locations, add them to reflect international customer preferences, and improve international marketing strategies. Similarly, AMT can help high-tech SMEs facilitate the connections and collaborations that are crucial for successful international integration. Therefore, we propose the following hypothesis:

Hypothesis 5b. AMT positively impacts IMA.

Technology adoption and its effects on a firm's bottom line are complex and multifaceted subjects that have been extensively investigated. The exact effect may vary depending on the type of technology, industry, and adoption stage [126,127]. However, noteworthily, the Metaverse is a relatively new and developing technology, and hence, several doubts regarding how it affects business outcomes still exist [128]. To take advantage of the benefits that it brings, firms must spend heavily on them. They may even have to change how they conduct business because each contributor has a narrative and perspective on several of Metaverse's greatest challenges and opportunities [54].

Businesses need to significantly increase customer engagement by providing immersive and interactive virtual experiences to improve satisfaction and retention. In the current research, by adopting Metaverse-related technologies, high-tech SMEs can improve their operational efficiency, growth rate, and new revenue streams, thereby gaining a competitive advantage on a global scale. However, research on this subject is lacking, and the effects may be unique to a particular market, business model, or adoption level. Therefore, in this study's context, we assume that AMT also may improve FP. Therefore, we hypothesize as follows:

Hypothesis 5c. AMT positively impacts FP.

3.2. Absorptive capacity's role

3.2.1. Absorptive capacity and IMA

Firm AbC refers to acquiring, absorbing, and using new knowledge and technology [79]. High AbC assists firms in identifying new

opportunities and adapting rapidly to technological innovation. This can boost their international success and competitive advantage [129]. However, inadequate AbC limits firms' adaptability and creativity, thus making global competition more challenging. Furthermore, it enables firms to strengthen ties with suppliers and partners, which is essential for flourishing in international marketplaces [84,130].

In the context of this study, a high-tech firm with a strong AbC is better positioned to spot emerging possibilities and adjust to shifting market conditions. As a firm can rapidly and efficiently adapt acquired knowledge to its marketing efforts, its responsiveness in this regard may increase. Assuming that a firm can use its AbC to create a distinctive operating competency such as agility, it will have a leg up on the competition [131], particularly in the international market in the Metaverse environment. Placing AbC in a Metaverse ecosystem highlights the unique contribution of this emerging technology to international marketing strategies, differentiating it from previous studies on AbC and performance. Furthermore, strong AbC can make it easier for a business to adjust to new Metaverse-related technologies, which is essential for expansion in the international economy through a virtual platform. This leads us to the following hypothesis:

Hypothesis 6a. Absorptive capacity positively impacts IMA.

3.2.2. Absorptive capacity and FP

A firm's ability to realize environmental changes directly impacts its performance because it enables the adoption of essential knowledge, the preservation of current knowledge and skills, and the most resilient management of available resources [83,132]. In highly competitive local, domestic, and international markets, a clear need exists for strategic flexibility to fulfill customers' changing needs efficiently, launch new products, adapt to negatively evolving political changes, build strategic partnerships, and deliver best-in-class services [40]. The literature suggests that AbC positively impacts FP. Kotabe et al. [133] concluded that AbC interacts with information acquisition to improve new products' market performance, which also may help increase marketing agility. Similarly, several studies have proposed that AbC increases a firm's performance in various ways [132,134].

High-tech firms that emphasize information acquisition and assimilation may bear the expenses of acquisition rather than the advantages of development despite their ability to refresh their knowledge base constantly. Firms must prioritize change; growth may reap temporary rewards, but they risk being caught in the "capacity trap." High AbC helps high-tech SMEs adjust to market changes, including AMT and industry upheavals, which can improve their performance over time. Therefore, we propose the following hypothesis:

Hypothesis 7a. Absorptive capacity positively impacts FP.

3.2.3. Absorptive capacity's mediating role

Flexible firms can better respond to shifting market trends and exploit opportunities than their competitors. Owing to high AbC, businesses can better comprehend, share, and integrate information such as that provided by customers, suppliers, and competitors [85,135]. Firms can gain a deeper understanding of new prospects than their rivals to effectively explore market potential and hazards [74]. A company with high AbC is better prepared to absorb new information, incorporate it into its current knowledge and practice, and use its resources to take advantage of national and international market trends [131]. Previous researchers have shown that AbC is a crucial dynamic capability connected to knowledge development and information utilization, which aid businesses in gaining and maintaining competitive advantage [78,136]. A firm's success results from its capacity to optimize operational possibilities and technology adoption to sustain a competitive advantage in response to rapid industry-wide and external change [137]. Based on this analysis of the relationships among AbC, FP, and AMT, this study infers that AbC plays an intermediary role in the relationships between AMT and IMA and between AMT and FP.

Hypothesis 6b. Absorptive capacity positively mediates the relationship between AMT and IMA.

Hypothesis 7b. Absorptive capacity positively mediates the relationship between AMT and FP.

3.3. IMA and FP

Agility is recognized as a vital aspect of facilitating firms' success in the complex business world, providing innovativeness and a competitive edge [39]. According to the literature, organizational agility refers to a firm's ability to adapt to rapid market change, including integrated strategies, paradigms, systems, and management practices based on multiple capabilities [138]. Organizational agility has garnered increasing attention from marketing researchers as a dynamic capacity influencing a firm's performance through standard marketing capabilities [139]. In this vein, Li et al. [140] described IMA as an organizational skill enabling firms to rapidly adopt marketing approaches in response to various local and foreign market scenarios. Based on the findings of Moi et al. [141], IMA represents a firm's ability to respond to and adapt marketing plans, establish intimate relationships from the outside in and the inside out, launch international marketing programs consistently and quickly, and predict and respond to future global market needs. Thoumrungroje and Racela [125] concluded that IMA is important to SMEs' adoption of new technologies and represents a key resource for improving SMEs' learning processes to identify and respond to market trends. Additionally, Gomes et al. [38] posited that IMA is essential for firms operating under multiple—even distant—sociocultural, economic, and performance challenges.

In this study's context, a high-tech firm that is exceptionally agile in its international marketing activities may respond rapidly to market shifts and gain new opportunities. In contrast, a less adaptable firm may need help to keep up with the rapid changes in the marketplace and lose lucrative opportunities. Furthermore, SMEs can adapt to international market changes more quickly because of their smaller size, leaner organizational structure, and decentralized decision making, which may enhance their performance. Previous research has largely ignored the fact that agility informs the international marketing perspective [36]. Further insight is needed to quantitatively assess IMA's impact on the performance of high-tech SMEs. Therefore, this study measures the effect of IMA on the performance of high-tech SMEs in emerging markets. Therefore, we propose the following hypothesis:

Hypothesis 8. IMA positively impacts FP.

Fig. 1 depicts this study's proposed research model and hypotheses.

4. Methodology

4.1. Context selection

To test the hypotheses of this study, we collected data from the executives of high-tech SMEs, such as software firms in Pakistan. The world's fifth most heavily populated country has opened several tech parks, allowing Pakistan's IT industry to "catapult boost" and double in size in 2 years. In addition to connecting with digital humans, the Metaverse can provide firms with internationalization opportunities. In FY21, Pakistan's total ICT exports grew by an average of 47 % per year to $2.1 billion, and they were expected to reach $3.5 billion by 2022 [142]. Meanwhile, the Metaverse market in Pakistan is expected to reach US$230.1 million by 2024, with an annual growth rate of 39.13 %, and to reach US$1669.0 million by 2030 [143]. The facts show that Pakistan is anticipating a high rate of Metaverse adoption across various sectors. The high-tech software industry may have the opportunity to reap the ultimate benefits of early AMT, increase sales, and gain international recognition.

The selection of high-tech software SMEs is deliberate because of

their creativity, economic clout, ability to drive technological breakthroughs, and forefront nature of innovation adoption. Moreover, Pakistani software firms often work with international customers and partners, making them more aware of global technology trends and more likely to adopt innovations like the Metaverse to stay competitive and meet international standards. Their smaller scale and agility allow them to develop and integrate creative solutions quickly, experiment with and adopt new technologies faster, and thus tap the potential of the Metaverse, set trends, and influence larger market dynamics. This may drive substantial economic growth among high-tech SMEs, possibly transforming them into larger market players. Although this industry faces unique challenges compared to other SMEs and large-scale firms in developing countries, the software industry is uniquely positioned to experiment with Metaverse-related technologies because of its agility and innovative mindset. Because of their unique qualities, this industry is ideally suited to investigate the revolutionary potential of Metaverse-related technologies, laying the groundwork for wider industry acceptance and integration of enterprise development.

Furthermore, most previous studies have focused solely on manufacturing SMEs and discussed ICT penetration rates; however, the software industry is also integral to economic development, especially in developing countries such as Pakistan. Therefore, this study's target population includes executives of various software firms intending to use innovative technology for business activities. The reason for selecting only executives as the sample is that they are the most prominent group in a firm and have a greater understanding of strategy, direction, and insights that can help firms better adopting Metaverse-related technologies and internationalization decisions.

4.2. Construct operationalization

A structured survey was administered to the target population to validate the proposed research paradigm. A 7-point Likert scale, with responses ranging from "1=strongly disagree" to "7=strongly agree," was used. All the measurement items were adopted from prior research and designed to meet this investigation's requirements (see the Appendix). Additionally, to evaluate the developed questionnaire's content validity, potential observations were analyzed separately by the authors. Once approved after discussion, the questionnaires were distributed. All constructs were measured using a lower-order construct, except AbC, which was calculated using a higher-order construct used by Cuevas-Vargas et al. [23], comprising four reflective dimensions: knowledge acquisition, assimilation of knowledge, transformation of knowledge, and exploitation of knowledge.

4.3. Sampling, data collection, and analysis

An online survey form was used for data collection through a computer-assisted web interview approach, a questionnaire data-gathering technique in which respondents use computers to complete surveys without being instructed by interviewers [144]. Firm profiles were derived from the Pakistan Software Export Board, which was created in 1995 as an apex government agency tasked with promoting Pakistan's IT industry in national and international markets [145]. The e-mails and contact numbers of >2000 firms were gathered and used for data analysis. We used a longitudinal design and divided our online survey into two phases. In the first phase, we used the information collected on exogenous variables (MI, CI, NI, OR, and PCM) and intention to adopt Metaverse-related technologies. A valid email address was required for survey participation, which helped filter out those who took the survey multiple times. This approach allowed us to track the process from intention to adoption. By addressing the temporal separation through a longitudinal study design, we provide a robust examination of how the intention to adopt the Metaverse translates into adoption behavior that leads to IMA achieving FP. A simple random sampling approach was used, and the survey was distributed to 600 firms in the first phase. This occurred during the first week of December 2021.

Of the 600 emails, 44 were not delivered successfully. Later, to obtain a higher response rate, we contacted the respondents via their contact numbers (provided by the Pakistan Software Export Board) through calls and text messages, encouraging them to respond to the survey. Before the start of this survey, we personally visited several firms, leveraging personal connections within the industry to build rapport and trust. Moreover, follow-up reminders were sent to non-respondents on different occasions to boost participation. By the end of March 2022, we had received only 280 responses. The survey filters ensured that the responses selected for analysis came from executives with some understanding of the Metaverse and related technologies. We asked about their understanding and involvement level of the Metaverse-related technologies and concepts (AR, VR, IoT, XR, blockchain, digital twins, AI, ML, NFT, and any others) using a 7-point Likert scale. In addition, we asked, "How often are you involved in projects or activities related to these technologies within your organization?" The results showed that the majority of responses were above average, and

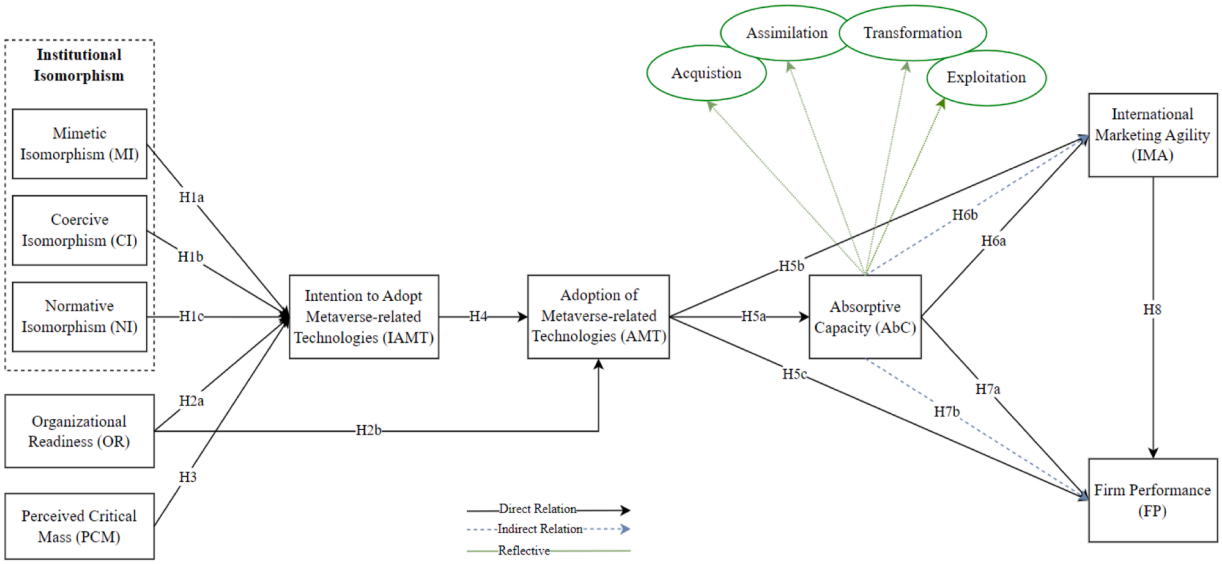


Fig. 1. Research Model.

the few below-average responses were excluded during the screening process. After screening, a total of 256 valid responses were recorded. In the second data collection phase between September 2022 and January 2023, we contacted our previous participants to obtain data on their AMT and the remaining study variables. We compared participants' emails from both phases to ensure that the same participants were included in the successive waves. Additionally, the respondents' demographic information was collected. Overall, 226 valid responses were received during the online survey's second phase. According to the literature, when performing structural equation modeling (SEM), a sample size of 200 or larger is appropriate [146]. Therefore, we considered it appropriate to perform the SEM analysis.

Based on the conclusions of Cohen's power theory [147], the appropriateness of the sample size was assessed. To assess the gathered sample's statistical strength, we used the post hoc G*power approach (with the significance threshold set at 0.05, an effect size of 0.15, and a sample size of 226) for all exogenous variables, including the reflecting indicators. The post hoc G*power results revealed that the statistical strength is significantly greater than the 0.8 criteria [147]. This quantitative study used a longitudinal design to evaluate causal hypotheses using partial least squares (PLS)-SEM in SmartPLS®4 statistical software [148,149]. PLS-SEM is considered a crucial technique for predictive analysis that helps evaluate theoretical frameworks [150]. PLS-SEM is widely accepted in social science and business research because of its elasticity and effectiveness in measuring the path values of complex models with multiple relationships [151]. Considering the complexity of the model and the relationship to be explored, to establish and test a new theoretical model, this study adopted PLS-SEM for analysis.

SmartPLS®4 provides nonparametric assessment measures based on "bootstrapping and blindfolding" because it lacks a distinctive goodness-of-fit criterion for evaluating findings. Of note, while using PLS-SEM, the measurement model was first estimated as a hierarchical components model using the indicator repetition technique, which is required to run higher-order models [148,152]. To further validate the results and overcome the limitations of PLS-SEM, we applied different robustness tests, including importance-performance map analysis (IPMA).

4.4. Method bias and multicollinearity

Some post hoc procedures have been assessed to demonstrate that the probable presence of common method bias (CMB) does not substantially alter interpretation of the data, ensuring that CMB is not a concern. First, Harman's single-factor test was used on all indicators in exploratory factor analysis using SPSS version 26 statistical software (IBM), with an unrotated first-factor accounting for only 43.93 % of the explained variance for all observed variables, including the dependent variable [153].

Second, when CMB was evaluated using a comprehensive collinearity assessment, neither the higher-order latent variables (ranging from 1.000 to 2.575) nor the lower-order latent variables (ranging from 1.019 to 3.169) passed the inner variance inflation factor (VIF) threshold value of 3.3 [154]. Third, the early and late responses from the data also were analyzed using the *t*-test, and no specific differences were found. These values were higher than the 5 % threshold suggested by previous studies [155]. Fourth, sequential randomization in the survey statements was applied to prevent systematic response patterns that also help avoid response bias.

Finally, we applied a nonperceptual marker variable to test the CMB issue. In longitudinal research, temporal separation of measurements can significantly reduce the risk of CMB. This is because data collected at different time points are less likely to be affected by the bias of the same respondent or by ephemeral factors that could influence responses simultaneously. We considered demographic variables (age, education, and job experience) as post hoc nonperceptual markers. We applied markers to all indicators in the model, and the results were not statistically significant in all cases, as shown in Table 3. The results were reliable because the relationship between the major constructs remained constant when the marker variables were included, which proves the validity of the first model finding. The results from these methods reveal that CMB had almost no effect on outcomes.

The outer VIF was used to check for multicollinearity. The highest VIF value discovered in this investigation was 2.575, well below the suggested threshold of 10 [156,157]. Consequently, no indication of considerable multicollinearity was observed.

5. Results

5.1. Demographics

Table 4 displays the respondents' demographic data, showing that nearly one-half (47.8 %) were young adults (those aged 30 and younger), and more than one-quarter (28.8 %) were in the 30-to-39-year-old bracket. Those in the 40 and older group accounted for 23.5 % of the total. Only 17.3 % of those surveyed were graduates, 68.5 % were postgraduates, and only 14.2 % had diplomas or other professional certifications qualifying them to enter or remain in the industry. Overall, 9.7 % of the total sample had between 1 and 3 years of experience, 42.9 % had between 4 and 6 years of experience, 22.6 % had between 7 and 10 years of experience, and 24.8 % had >10 years of experience. These findings show that the high-tech software industry has a relatively young and highly educated workforce. The software industry is characterized by a highly experienced workforce, which is exemplified by the massive influx of highly trained newcomers who contribute to developing novel approaches and actively participate in the software industry's innovation loop.

5.2. Reliability and convergent validity

Table 5 presents the results on convergent validity and reliability. The composite reliability readily exceeded the essential suggested value of 0.7, indicating that lower- and higher-order constructions have excellent internal consistency [151]. Moreover, Cronbach's alpha for each construct exceeded 0.7 (with values ranging from 0.785 to 0.929). Furthermore, for each construct, the rho_A significance level exceeded 0.7 (ranging from 0.796 to 0.929), as suggested by Dijkstra et al. [158]. Meanwhile, all constructs meet the actual threshold of 0.5 of the average variance extracted (AVE) [159,160]. Additionally, all indicators' outer loadings exceeded 0.7 [151], as evidenced by their ranges; all were statistically significant ($p < 0.001$), ensuring the heterogeneity of each manifested variable. To determine an organization's AbC, this study

Table 3
Marker variable outcomes.

Path	Standardized Coefficient	S.D.	t-values	p-values
Marker Variable → AMT	−0.037	0.079	0.467	0.640
Marker Variable → AbC	−0.048	0.088	0.544	0.587
Marker Variable → CI	−0.122	0.120	1.014	0.311
Marker Variable → FP	−0.033	0.047	0.697	0.486
Marker Variable → IAMT	−0.035	0.049	0.720	0.471
Marker Variable → IMA	−0.078	0.071	1.087	0.277
Marker Variable → MI	−0.081	0.098	0.832	0.405
Marker Variable → NI	0.052	0.097	0.540	0.589
Marker Variable → OR	0.017	0.095	0.183	0.855
Marker Variable → PCM	−0.003	0.076	0.045	0.964

Adoption of Metaverse-related technologies (AMT), Coercive Isomorphism (CI), Firm Performance (FP), International Marketing Agility (IMA), Intention to adopt Metaverse-related technologies (IAMT), Mimetic Isomorphism (MI), Normative Isomorphism (NI), Organizational Readiness (OR), Perceived Critical Mass (PCM), Absorptive Capacity (AbC).

examined how well it functions across four key categories. The results for the higher-order construct reveal a significant relationship in the reflective model. The results demonstrate that convergent validity and reliability are of no concern.

5.3. Discriminant validity

Discriminant validity is a well-known requirement for investigating connections among latent components. We used three methods to test discriminant validity: the Heterotrait-Monotrait ratio (HTMT), Fornell-Larcker criterion (presented in Table 6), and the cross-loading criterion [152,161,162]. First, the HTMT correlation ratio findings are presented above the diagonal (boldface) values, wherein the values of the correlations between the reflective constructs were <0.933 after thorough bootstrapping [162]. Second, the Fornell-Larcker criteria test was applied, taking the square root of the AVE for each construct, the values of which are presented below the diagonal (boldface) values in Table 6. According to Ab Hamid et al. [161], these values are greater than their comparable associations with any other construct, as presented on the diagonal.

Third, construct cross-loadings were used to determine the discriminant validity [162]. Each item's loading must be greater than the loadings of its following constructs. Table 7 indicates that the findings meet the cross-loading requirements. Based on these findings, it can be inferred that the data from this study are dependable and valid for testing the research hypotheses.

5.4. The structural model

We used PLS-SEM to confirm the study model's dependability, convergence, and discriminant validity. Fig. 2 depicts the path coefficient values, which are also listed in Table 11. The SEM route's significance level was established using bootstrapping with 5000 resamplings.

The results' explanatory deviation quantifies the proposed model's descriptive capability (i.e., the R^2 value). The results show an R^2 value of 0.598 for IAMT, indicating that the selected factors explain 59.8 % of the variation. The R^2 value of AMT of 0.228 indicates a 22.8 % variation. Similarly, the R^2 of 0.229 for AbC indicates a 22.9 % variation. The R^2 value of IMA was 0.502, and the R^2 value of FP was 0.608, suggesting 50.2 % and 60.8 % variation, respectively, from the latent construct in the path model. The results in Table 8 allow us to infer that IAMT, AMT, IMA, AbC, and FP have explanatory capacities, as recommended previously [163].

The predictive relevance of the model's endogenous constructs was examined using the Stone-Geisser blindfolding approach [164,165]; the results are presented in Table 8. A Q^2 value exceeding 0 for a specific endogenous reflective construct indicates the path model's predictive importance [152]. In this study, the Q^2 values of IAMT, AMT, AbC, IMA, and FP were 0.394, 0.186, 0.165, 0.335, and 0.430, respectively, in their capacity as endogenous variables, demonstrating their predictive importance owing to Q^2 values greater than 0. Consequently, this model is predictive of all endogenous components.

Table 4
Respondent demographics.

Category		Frequency	Percent
Age	20–29 years	108	47.8
	30–39 years	65	28.8
	Above 40	53	23.5
	Total	**226**	**100.0**
Education	Graduate	39	17.3
	Post-Graduate	155	68.5
	Diploma/Professional certification	32	14.2
	Total	**226**	**100.0**
Job Experience	1–3 years	22	9.7
	4–6 years	97	42.9
	7–10 years	51	22.6
	> 10 years	56	24.8
	Total	**226**	**100.0**

Table 9 presents the results of PLS-predict and cross-validated predictive ability test (CVPAT) analysis. Q^2predict values are used to measure the predictive relevance of the model, and values greater than 0 indicate that the model has predictive relevance. The results show that the Q^2predict values of all constructs are positive, indicating that the predictive ability of the model for these constructs is satisfactory. Meanwhile, values of root mean square error and mean absolute error also were low, indicating good model performance.

The CVPAT also was used to evaluate and verify the predictive ability of the model [166]. The results show that the average loss difference of all constructs was negatively significant, indicating that the PLS-SEM model can predict the results better than the simple model. The overall average loss difference was −0.289 ($p < 0.001$), further confirming that the PLS-SEM model is generally better at predicting outcomes than the simple model.

The results in Table 10 show a correlation matrix between studied constructs, and each correlation value represents the direction and strength of the linear connection between constructs. All the values show strong positive correlations between the constructs.

Table 11 includes the beta coefficients for bootstrapping and significant values of the direct and indirect influences. As shown in the results, the t-statistic and p-value are higher than the suggested values [152,167]. The structural model in Fig. 2 reveals that 13 paths are significantly accommodating. Table 11 indicates that MI (β=0.135, $p < 0.05$), CI (β=0.271, $p < 0.01$), NI (β=0.200, $p < 0.05$), OR (β=0.156, $p < 0.05$), and PCM (β=0.177, $p < 0.01$) exhibit significant effects on IAMT. Similarly, OR positively impacts AMT (β=0.341, $p < 0.001$), and IAMT is also positively related to AMT (β=0.185, $p < 0.01$). Meanwhile, AMT significantly impacts AbC (β=0.479, $p < 0.001$), IMA (β=0.216, $p < 0.001$), and FP (β=0.151, $p < 0.01$). Furthermore, the results reveal that FP is significantly influenced by the AbC (β=0.363, $p < 0.001$) and IMA (β=0.388, $p < 0.001$). Meanwhile, the IMA is significantly influenced by AbC (β=0.579, $p < 0.001$).

Furthermore, the indirect effect of AMT → AbC → IMA (β=0.277, $p < 0.001$) and AMT → AbC → FP (β=0.174, $p < 0.001$) is also statistically significant. Our findings empirically support AbC's mediating role in our theoretical model. Higher AMT levels directly increase IMA and FP, but also increase AbC, thus leading to IMA and FP. As both the indirect and direct effects are significant and positive, they are regarded as partial mediation [168]. The mediation results are presented in Table 12.

5.5. IPMA

PLS-SEM introduces a novel IPMA approach, which calculates the performance of every construct to offer a better understanding of the PLS-SEM results. IPMA benefits by estimating path coefficients or importance metrics and analyzing the underlying configuration and average of the related performance metrics [169]. The IPMA approach is based on the idea that the total effect represents a given factor's influence on the target factor and its importance, wherein construct performance is assessed using the mean of the underlying construct [5,169]. IPMA, also known as "prioritization analysis, importance-performance matrix, or impact-performance map, expands the results of path coefficients (importance) by integrating the average of potential construct scores (performance)" [169]. It compares the unstandardized overall impact (importance) in the structural model to the latent structure scores' mean on a performance scale ranging from 0 % to 100 %. IPMA analysis was performed only on endogenous constructs, to analyze and compare all relevant exogenous constructions.

Fig. 3 presents the IPMA values for FP. The mean value of importance was 0.203, and that of performance was 62.396 %. Regarding the priority improvement in FP, AMT, and AbC should be given paramount priority, followed by AbC, AMT, CI, IMA, IAMT, MI, NI, OR, and PCM.

Fig. 4 illustrates the IPMA values for IMA. The mean value of

Table 5
Measurement model (reflective model) results.

Lower-Order Constructs	Internal Consistency					
	Indicators	Outer Loading Range	Cronbach's Alpha	rho_A	CR	AVE
AC	AC1, AC2, AC3	0.844–0.891	0.847	0.851	0.908	0.766
AMT	AMT1, AMT2, AMT3	0.932–0.940	0.929	0.929	0.955	0.875
AS	AS1, AS2, AS3, AS4	0.824–0.885	0.873	0.875	0.913	0.724
CI	CI1, CI2, CI3	0.861–0.925	0.883	0.889	0.928	0.811
EX	EX1, EX2, EX3	0.864–0.908	0.865	0.870	0.917	0.787
FP	FP1, FP2, FP3, FP4, FP5	0.802–0.875	0.903	0.904	0.928	0.722
IMA	IMA1, IMA2, IMA3, IMA4	0.803–0.879	0.846	0.847	0.897	0.684
IAMT	IAMT1, IAMT2, IAMT3	0.720–0.900	0.785	0.796	0.877	0.706
MI	MI1, MI2, MI3, MI4	0.747–0.904	0.869	0.891	0.911	0.719
NI	NI1, NI2, NI3	0.812–0.889	0.822	0.825	0.894	0.738
OR	OR1, OR2, OR3, OR4	0.847–0.859	0.876	0.882	0.914	0.727
PCM	PCM1, PCM2, PCM3	0.780–0.928	0.852	0.873	0.911	0.775
TR	TR1, TR2, TR3, TR4	0.753–0.892	0.866	0.869	0.910	0.717

Higher-Order Construct	Variables	Path coefficients (t-value)	Cronbach's Alpha	rho_A	CR	AVE
Absorptive Capacity	AC	0.853 (26.149)	0.892	0.897	0.925	0.756
	AS	0.903 (47.908)				
	EX	0.842 (30.611)				
	TR	0.879 (37.668)				

Note: The t-values of all outer loadings were significant ($p < 0.001$). Abbreviations: Acquisition (AC), Adoption of Metaverse-related technologies (AMT), Assimilation (AS), Composite reliability (CR), Coercive Isomorphism (CI), Exploitation (EX), Firm Performance (FP), International Marketing Agility (IMA), Intention to adopt Metaverse-related technologies (IAMT), Mimetic Isomorphism (MI), Normative Isomorphism (NI), Organizational Readiness (OR), Perceived Critical Mass (PCM), Transformation (TR).

Table 6
Discriminant validity for lower-order constructs.

Constructs	AC	AMT	AS	CI	EX	FP	IMA	IAMT	MI	NI	OR	PCM	TR
AC	**0.875**	0.413	0.838	0.571	0.716	0.692	0.698	0.519	0.665	0.645	0.639	0.540	0.744
AMT	0.368	**0.936**	0.529	0.471	0.440	0.564	0.556	0.461	0.492	0.441	0.499	0.332	0.466
AS	0.721	0.479	**0.851**	0.621	0.762	0.738	0.729	0.544	0.655	0.679	0.666	0.571	0.845
CI	0.499	0.429	0.547	**0.900**	0.615	0.646	0.686	0.827	0.656	0.837	0.750	0.606	0.626
EX	0.613	0.396	0.663	0.541	**0.887**	0.624	0.622	0.533	0.502	0.571	0.626	0.456	0.793
FP	0.606	0.517	0.656	0.577	0.552	**0.850**	0.813	0.680	0.680	0.683	0.680	0.569	0.697
IMA	0.594	0.494	0.629	0.594	0.533	0.710	**0.827**	0.638	0.685	0.741	0.683	0.551	0.719
IAMT	0.429	0.392	0.454	0.688	0.443	0.572	0.519	**0.840**	0.697	0.839	0.721	0.713	0.569
MI	0.565	0.440	0.567	0.576	0.430	0.601	0.588	0.584	**0.848**	0.683	0.580	0.639	0.607
NI	0.543	0.385	0.579	0.715	0.484	0.591	0.620	0.673	0.580	**0.859**	0.724	0.708	0.656
OR	0.553	0.454	0.586	0.663	0.551	0.608	0.593	0.607	0.509	0.620	**0.852**	0.506	0.667
PCM	0.461	0.297	0.493	0.528	0.393	0.502	0.470	0.582	0.552	0.592	0.447	**0.880**	0.517
TR	0.634	0.417	0.733	0.546	0.683	0.617	0.615	0.467	0.526	0.552	0.583	0.443	**0.847**

Note: The diagonal numbers (bold-faced) represent the square root of the AVE (for reflective constructs). Below the diagonal (bold-faced), the Fornell-Larcker criterion is presented; above the diagonal (bold-faced), the HTMT ratio test is presented.

importance was 0.174, and that of performance was 62.168 %. Based on these findings, AMT should be given the highest priority for improvement in the IMA, followed by AbC, CI, IAMT, MI, NI, OR, and PCM.

Fig. 5 depicts the IPMA values of the AMT. The mean values of importance and performance are 0.113 and 62.002%, respectively. To improve the AMT, priority should be given to CI, followed by IAMT, MI, NI, OR, and PCM.

Similarly, Fig. 6 shows that mean values of importance and performance for the IAMT of 0.188 and 62.142 %, respectively. To improve the IAMT, CI should be prioritized, followed by MI, NI, OR, and PCM.

5.6. Age segmentation

We separated the data into three categories based on the age of the respondents to run the PLS-MGA. Category 1 consists of respondents age 20–30 years; category 2, of respondents age 31–40 years; and category 3, of respondents age 41 years and older. Furthermore, to compare the results of these 3 age categories, we created 3 groups, denoted A, B, and C. Group A compares the results of the age categories 1 and 2; group B, the results of age categories 1 and 3; and group C compares the results of age categories 2 and 3.

First, we computed the results using PLS-MGA for group A; the results are shown in Table 13. Next, we computed the results for group B, which revealed no difference in the significance and direction of the relationship in the age groups. Finally, we computed the results for group C, which also demonstrated no significant difference the relationships among the age groups.

In comparing the age groups A, B, and C, the aim was to check the differences in the adoption decisions of Metaverse-related technologies in high-tech SMEs among young, middle-aged, and older executives. There is a lack of representation of such cases in literature, which was confusing when interpreting the results. Our PLS-MGA comparison analysis showed no statistically significant difference among these three age groups and confirmed the cross-age stability of the results. Therefore, our study shows that all age groups consider Metaverse-related technologies useful in business.

6. Discussion

Our results provide significant support for all our proposed hypotheses, demonstrating that MI positively impacts IAMT. These results suggest that SMEs' IAMT can be generated by imitating other firms' behavior. For nonadopters, imitating competitors is a reliable means to motivate them to create their IAMT. Similarly, CI positively impacts

Table 7
Item Cross-loadings.

Items	AC	AMT	AS	CI	EX	FP	IMA	IAMT	MI	NI	OR	PCM	TR
AC1	**0.890**	0.366	0.611	0.518	0.527	0.545	0.563	0.453	0.509	0.542	0.492	0.460	0.490
AC2	**0.891**	0.221	0.608	0.426	0.522	0.508	0.511	0.367	0.448	0.461	0.436	0.388	0.534
AC3	**0.844**	0.319	0.619	0.339	0.528	0.543	0.456	0.310	0.501	0.408	0.497	0.364	0.561
AMT1	0.342	**0.939**	0.430	0.396	0.352	0.439	0.450	0.333	0.369	0.325	0.428	0.243	0.376
AMT2	0.350	**0.933**	0.440	0.390	0.400	0.479	0.444	0.378	0.418	0.389	0.391	0.302	0.413
AMT3	0.343	**0.927**	0.450	0.401	0.334	0.482	0.455	0.354	0.413	0.344	0.416	0.258	0.362
AS1	0.602	0.444	**0.848**	0.486	0.537	0.582	0.553	0.396	0.478	0.513	0.458	0.426	0.529
AS2	0.610	0.403	**0.886**	0.491	0.566	0.555	0.544	0.428	0.486	0.511	0.491	0.437	0.607
AS3	0.569	0.326	**0.822**	0.395	0.549	0.519	0.468	0.310	0.421	0.408	0.498	0.392	0.631
AS4	0.671	0.419	**0.839**	0.447	0.564	0.539	0.548	0.389	0.528	0.507	0.544	0.416	0.706
CI1	0.419	0.321	0.464	**0.860**	0.413	0.481	0.489	0.568	0.497	0.630	0.544	0.454	0.452
CI2	0.461	0.405	0.495	**0.924**	0.540	0.546	0.583	0.624	0.529	0.661	0.620	0.492	0.517
CI3	0.465	0.413	0.491	**0.912**	0.472	0.506	0.522	0.657	0.512	0.630	0.609	0.470	0.478
EX1	0.550	0.311	0.542	0.461	**0.856**	0.444	0.450	0.352	0.378	0.399	0.449	0.343	0.598
EX2	0.569	0.368	0.623	0.510	**0.906**	0.522	0.506	0.408	0.378	0.449	0.497	0.328	0.573
EX3	0.512	0.346	0.557	0.428	**0.886**	0.473	0.435	0.391	0.381	0.425	0.490	0.354	0.592
FP1	0.518	0.475	0.553	0.497	0.444	**0.875**	0.605	0.495	0.557	0.528	0.550	0.460	0.527
FP2	0.473	0.457	0.567	0.482	0.456	**0.857**	0.578	0.462	0.519	0.491	0.490	0.401	0.474
FP3	0.573	0.424	0.576	0.463	0.456	**0.834**	0.572	0.473	0.532	0.502	0.494	0.418	0.532
FP4	0.529	0.366	0.541	0.491	0.482	**0.866**	0.635	0.497	0.487	0.527	0.522	0.433	0.541
FP5	0.479	0.396	0.503	0.476	0.471	**0.802**	0.604	0.492	0.436	0.443	0.500	0.428	0.452
IMA1	0.539	0.434	0.531	0.524	0.396	0.595	**0.834**	0.407	0.508	0.504	0.507	0.394	0.503
IMA2	0.491	0.381	0.517	0.472	0.436	0.600	**0.838**	0.469	0.515	0.522	0.481	0.448	0.503
IMA3	0.463	0.429	0.532	0.510	0.468	0.536	**0.809**	0.417	0.448	0.503	0.521	0.341	0.528
IMA4	0.469	0.342	0.473	0.442	0.437	0.598	**0.813**	0.423	0.467	0.508	0.456	0.365	0.458
IAMT1	0.278	0.303	0.294	0.540	0.261	0.438	0.395	**0.718**	0.396	0.525	0.434	0.492	0.343
IAMT2	0.358	0.329	0.366	0.565	0.388	0.495	0.438	**0.888**	0.520	0.571	0.516	0.499	0.382
IAMT3	0.435	0.322	0.459	0.615	0.431	0.496	0.468	**0.889**	0.530	0.590	0.564	0.478	0.415
MI1	0.496	0.331	0.474	0.416	0.379	0.476	0.426	0.385	**0.736**	0.434	0.412	0.346	0.422
MI2	0.499	0.384	0.491	0.484	0.336	0.514	0.525	0.523	**0.901**	0.486	0.421	0.514	0.462
MI3	0.463	0.348	0.485	0.533	0.375	0.527	0.509	0.574	**0.889**	0.547	0.470	0.497	0.463
MI4	0.478	0.393	0.466	0.482	0.372	0.509	0.522	0.452	**0.839**	0.478	0.426	0.476	0.406
NI1	0.410	0.297	0.455	0.562	0.364	0.464	0.480	0.551	0.457	**0.808**	0.450	0.504	0.446
NI2	0.487	0.346	0.456	0.571	0.405	0.483	0.525	0.557	0.486	**0.877**	0.503	0.499	0.456
NI3	0.497	0.325	0.551	0.682	0.459	0.556	0.573	0.615	0.535	**0.875**	0.620	0.514	0.478
OR1	0.455	0.334	0.477	0.542	0.414	0.481	0.471	0.452	0.407	0.507	**0.857**	0.325	0.488
OR2	0.468	0.364	0.475	0.529	0.432	0.510	0.482	0.430	0.389	0.489	**0.857**	0.342	0.467
OR3	0.493	0.420	0.528	0.563	0.517	0.544	0.547	0.599	0.457	0.517	**0.847**	0.420	0.508
OR4	0.463	0.373	0.505	0.602	0.469	0.514	0.522	0.556	0.475	0.586	**0.848**	0.418	0.499
PCM1	0.371	0.200	0.386	0.393	0.270	0.365	0.353	0.435	0.462	0.471	0.306	**0.779**	0.328
PCM2	0.439	0.293	0.452	0.489	0.404	0.493	0.461	0.538	0.490	0.556	0.436	**0.915**	0.410
PCM3	0.406	0.255	0.454	0.492	0.332	0.462	0.416	0.556	0.494	0.528	0.422	**0.927**	0.388
TR1	0.537	0.330	0.630	0.473	0.553	0.517	0.534	0.390	0.494	0.464	0.488	0.364	**0.894**
TR2	0.531	0.350	0.570	0.448	0.560	0.452	0.505	0.383	0.443	0.450	0.440	0.401	**0.750**
TR3	0.534	0.356	0.622	0.455	0.550	0.531	0.503	0.340	0.409	0.425	0.492	0.311	**0.885**
TR4	0.546	0.358	0.636	0.443	0.588	0.519	0.507	0.433	0.413	0.488	0.532	0.384	**0.851**

Note. All factor loadings are significant at the $p < 0.001$ level. Boldface values represent the item loadings for each factor.

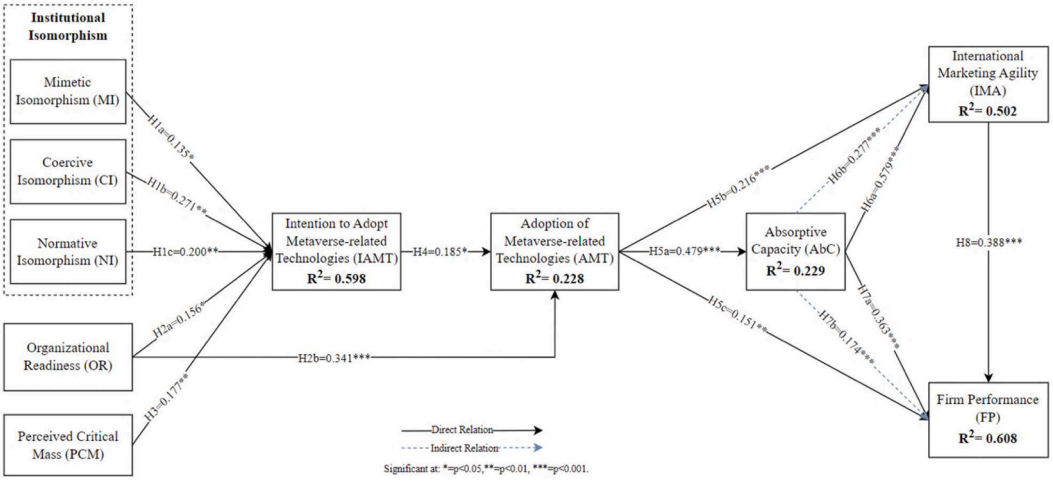


Fig. 2. SEM results.

Table 8
Construct cross-validated redundancy.

Constructs	SSO	SSE	Q^2 (=1-SSE/SSO)	R^2
IAMT	678.000	410.661	0.394	0.598
AMT	678.000	552.046	0.186	0.228
AbC	904.000	754.564	0.165	0.229
IMA	904.000	600.741	0.335	0.502
FP	1130.000	644.407	0.430	0.608

Table 9
Predictive relevance and CVPAT.

Constructs	Prediction results			CVPAT (PLS-SEM vs. Indicator average)		
	Q^2predict	RMSE	MAE	Average loss difference	t value	p-value
AMT	0.223	0.894	0.673	−0.317	2.686	0.008
AbC	0.254	0.883	0.594	−0.193	4.268	0.000
FP	0.250	0.882	0.660	−0.310	4.393	0.000
IMA	0.237	0.889	0.634	−0.241	4.205	0.000
ITAM	0.564	0.673	0.497	−0.422	3.983	0.000
Overall	—	—	—	−0.289	4.652	0.000

SMEs' IAMT. These results indicate that leading firms incorporate protocols to manage resources, data, and Metaverse technology adoption environments. Businesses that neglect these protocols and guidelines may experience consequences that severely impact their performance and reputation. In high-tech industries, customers invariably demand a high level of technological innovation, and firms in the industry react to avoid losing their competitive edge [94]. Regarding Metaverse adoption in high-tech SMEs, the results indicate that firms with a high degree of CI are more likely to adopt Metaverse-related technologies to maintain a dominant market position. Furthermore, NI positively impacts SMEs' IAMT. The study respondents indicated that firms felt strained to adhere to the normative patterns driving IAMT in the high-tech industry. In this situation, the firm may not correctly assess the costs, advantages, and hazards of Metaverse-related technologies, resulting in unsatisfactory outcomes. Sources of NI can include professional bodies, industry standards, stakeholders, the media, and other agencies promoting SMEs' IAMT. The findings of this study can be integrated with those of other studies to provide a broader picture of how high levels of institutional isomorphism (MI, CI, and NI) positively impact higher intention to adopt technology [20,94]. Therefore, the study findings support H1a, H1b, and H1c.

Similarly, OR positively impacts SMEs' IAMT. Based on these results, we conclude that high-tech SMEs with greater dedication, resources, and understanding are more likely to adopt Metaverse-related technologies. Furthermore, adequate financing arguably allows high-tech SMEs to build strategies and processes, invest in technology, provide training in the ability and knowledge to implement Metaverse-related technologies, and obtain the assistance required to lead their IAMT. Therefore, based on these results, we can assume that financial and technological competence reflects a major OR that aids in creating IAMT, leading to AMT in high-tech SMEs in developing countries. These outcomes are consistent with those of previous studies finding that OR has a beneficial influence on the adoption of innovative technologies [94,170]. Thus, based on this discussion, H2a and H2b are supported.

Table 11
Structural equation modeling analysis.

Hypotheses	Standardized Coefficient	S.D.	t-values	p-values
H1a= MI → IAMT	0.135*	0.063	2.133	0.033
H1b= CI → IAMT	0.271**	0.090	3.022	0.003
H1c= NI → IAMT	0.200**	0.084	2.381	0.017
H2a= OR → IAMT	0.156*	0.077	2.027	0.043
H2b= OR → AMT	0.341***	0.091	3.768	0.000
H3= PCM → IAMT	0.177**	0.067	2.653	0.008
H4= IAMT → AMT	0.185*	0.078	2.367	0.018
H5a= AMT → AbC	0.479***	0.064	7.498	0.000
H5b= AMT → IMA	0.216***	0.063	3.425	0.001
H5c= AMT → FP	0.151**	0.051	2.983	0.003
H6a= AbC → IMA	0.579***	0.074	7.815	0.000
H7a= AbC → FP	0.363***	0.061	5.986	0.000
H8= IMA → FP	0.388***	0.056	6.897	0.000

Significance values: ***=$p < 0.001$, **=$p < 0.01$. *=$p < 0.05$.

Our results indicate that PCM positively influences IAMT in high-tech SMEs. These findings further demonstrate the significance of PCM for understanding the dynamics of IAMT in high-tech SMEs in developing countries. Adoption intention is influenced by whether industry peers adopt the same technology. Firms that believe that other industry members are adopting Metaverse-related technologies are more likely to adopt them as well. These results contribute to similar findings in various contexts [32,33]; therefore, H3 is also supported. A sizeable technological shift can dramatically remodel the industry structure and define the framework of SMEs' competitive tactics to gain a new competitive advantage. This technological shift ensures the delivery of highly realistic and low-latency VR experiences to users in the virtual world. This has enhanced the Metaverse's reputation and facilitated the conversion of intention toward adoption by high-tech firms. Furthermore, this study confirms the strong impact of IAMT on AMT in high-tech SMEs. Therefore, H4 is supported.

Our results confirm that allocating the resources and capabilities available for innovation and the right action dimension with proper knowledge would allow internal and external information to better adapt to a firm's innovation needs. Therefore, high-tech SMEs must develop capacities that enable them to deliver services locally and internationally. Previous studies have demonstrated that innovation activities and the adoption of novel technologies positively impact performance [132]. The results of this study extend prior findings by arguing for the positive impact of AMT on IMA and FP. Meanwhile, AMT benefits knowledge-processing capabilities, enhancing FP. Therefore, the study findings support H5a, H5b, and H5c.

One of the study's most interesting results is that the AbC of knowledge enables high-tech SMEs to expand their international marketing operations and boosts their business success. This suggests that

Table 10
Constructs correlations.

Constructs	AMT	AbC	CI	FP	IMA	ITAM	MI	NI	OR	PCM
AMT	1.000									
AbC	0.479	1.000								
CI	0.429	0.613	1.000							
FP	0.517	0.700	0.577	1.000						
IMA	0.494	0.683	0.593	0.710	1.000					
ITAM	0.393	0.515	0.689	0.572	0.519	1.000				
MI	0.440	0.603	0.576	0.601	0.588	0.584	1.000			
NI	0.386	0.622	0.715	0.591	0.619	0.673	0.580	1.000		
OR	0.454	0.654	0.663	0.608	0.593	0.607	0.509	0.620	1.000	
PCM	0.297	0.516	0.528	0.502	0.470	0.583	0.552	0.592	0.447	1.000

Table 12
Mediation results.

Hypotheses	Path a	Path b	Indirect Effect	SE	t-value	Bootstrapped Confidence Interval at 95 %	
						Lower limit	Upper limit
H6b= AMT → AbC → IMA	0.479	0.579	0.277	0.055	4.996	0.169	0.386
H7b= AMT → AbC → FP	0.479	0.363	0.174	0.039	4.439	0.097	0.251

Note: Path values are statistically significant at $p < 0.001$.

the activities of high-tech SMEs—such as enhanced intellectual capital management and transformation—are essential for operating virtual communication platforms. This study shows that AbC exerts a significant and positive influence on FP, which is consistent with Kale et al. [40]. Furthermore, research has demonstrated that the capacity to acquire, assimilate, transfer, and develop information is critical for marketing operations because it has both direct and indirect consequences for increased IMA and FP. The outcomes of knowledge acquisition, assimilation, transformation, and exploitation are conducive to a developing firm's AbC through incoming information management; notably, these outcomes were consistent [75]. Therefore, the study findings support H6a, H6b, H7a, and H7b.

Moreover, this study measured the effect of IMA on FP and demonstrated that IMA exhibits a significant positive impact on FP. Notably, firms need marketing agility and participation in adjusting international marketing programs to remain competitive. The results reveal that the effect of IMA is more significant when a firm adopts an adaptive marketing strategy [37]. An increase in high-tech SMEs' IMA may increase their FP. Therefore, H8 is also supported.

This study helps understand the perception of executives of different age groups (from young to old) of the value of Metaverse-related technologies in high-tech SMEs. The findings suggest that age does not play a significant role in the decisions to adopt these technologies, highlighting the potential universality of these technologies across different age groups. This research validates the consistency of the findings across age groups and demonstrates that participants of all ages find Metaverse-related technologies valuable in business.

6.1. *Theoretical contributions and implications*

This study makes significant contributions to academic knowledge in the following ways First, it fills a gap in the literature on high-tech SMEs by demonstrating that IAMT is strongly influenced by the factors identified in this research. The study contributes to the literature on technology adoption by shedding light on the factors influencing SMEs' IAMT. This new perspective has prompted a reevaluation of current technology adoption theories and implies that existing models should be modified by considering the unique characteristics and strategic goals of smaller, more agile enterprises. This can be important as global technology innovations, such as the Metaverse and platforms, continue disrupting high-tech SMEs.

Second, previous studies on innovative technology adoption have

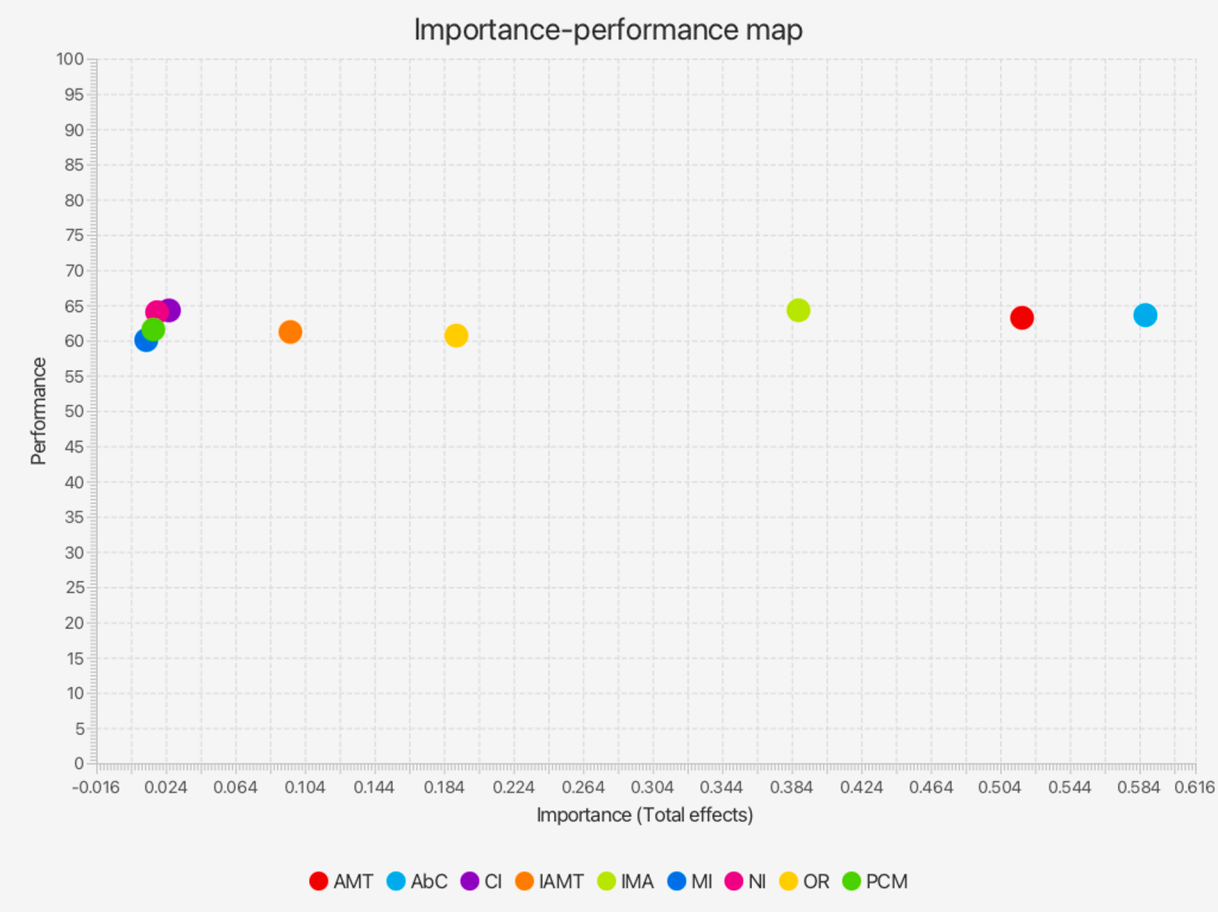


Fig. 3. Importance-performance map analysis for FP.

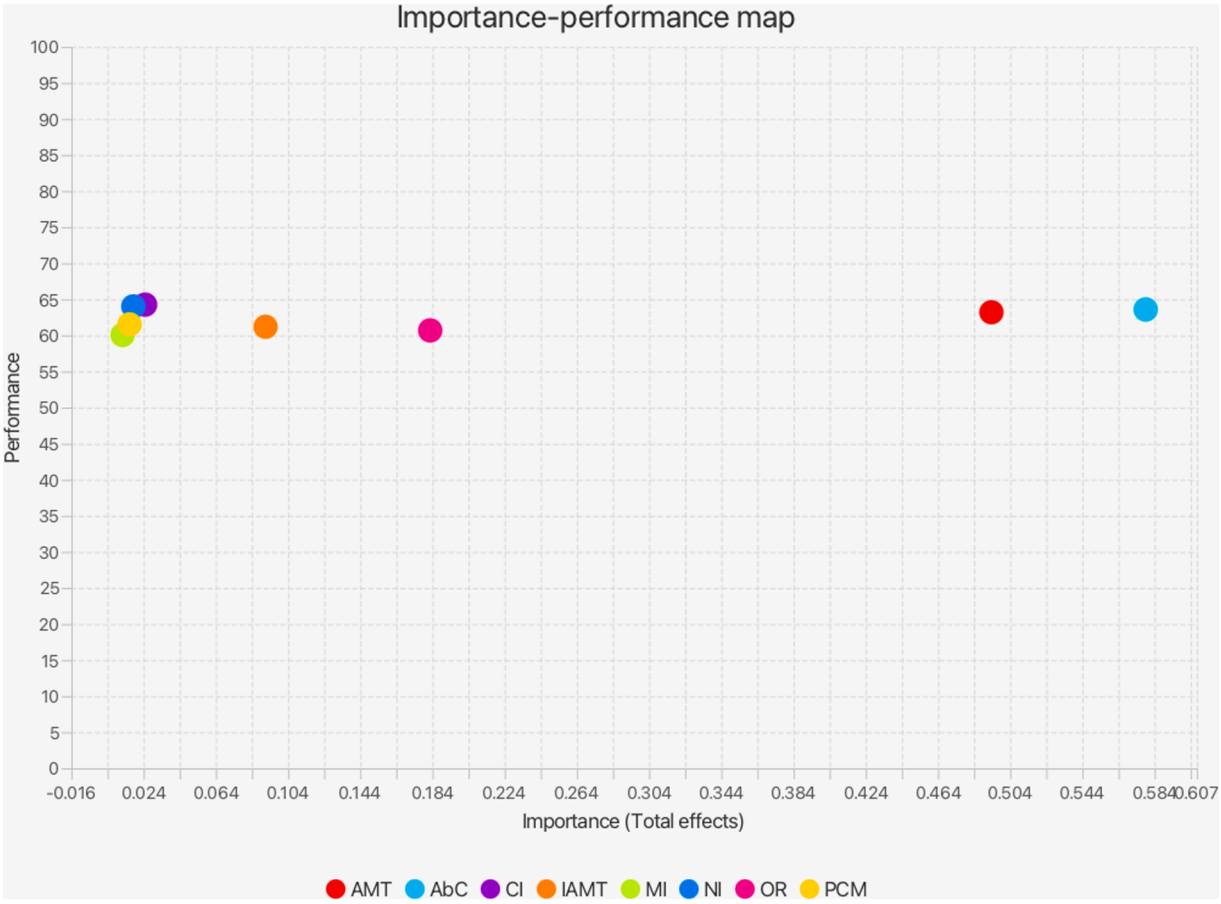


Fig. 4. Importance-performance map analysis for IMA.

typically examined large-scale industries and other government organizations; however, SMEs have unique characteristics that differ from those of large-scale enterprises. Therefore, this study fills a gap in the literature on adopting innovative technology by focusing on a hitherto underexplored population—namely, high-tech SMEs. Hence, this study provides a novel perspective on AMT and a better understanding of the various aspects and concerns that high-tech SMEs face, particularly in developing countries. This offers a new perspective on technology adoption by smaller, more agile enterprises in the upcoming virtual world. It also broadens the scope of the technology adoption literature and provides valuable insights into the strategic decision-making process of SMEs. Thus, this fills a critical gap in understanding how high-tech SMEs interact with cutting-edge technologies related to the development of the Metaverse ecosystem.

Third, this research adds to the existing body of knowledge in institutional theory by pinpointing the precise mechanism by which institutional pressures (i.e., MI, NI, and CI) affect AMT in SMEs. The study demonstrates that MI and NI significantly and positively impact IAMT, and CI is the stronger of these two factors (because of its higher value in the path model). This suggests that institutional forces, which are often studied in more developed environments, play a key role in the adoption of new and rapidly evolving technologies, such as the Metaverse. Moreover, emphasis is placed on how this study integrates elements such as OR and PCM to provide a more nuanced knowledge of institutional theory that can improve on existing theoretical frameworks. For example, according to the study, high-tech SMEs are more likely to adopt technologies associated with the Metaverse ecosystem when they believe that many of their competitors are already using these technologies. Similarly, SMEs that are better prepared organizationally—that is, have the required staff and infrastructure—are better able to adapt to the normative constraints imposed by industrial codes and standards. This may help extend institutional theory's horizons by providing a more comprehensive knowledge of the factors driving AMT by SMEs and enhance institutional theory by incorporating these variables.

Finally, this study contributes to the international business and marketing literature by presenting IMA as an alternative method for evaluating high-tech SMEs' dynamic capability to respond to changes in the international market and achieve performance goals. The model provides a robust alternative method for assessing how SMEs can capitalize on organizational capacity and adapt to novel technological developments (such as the Metaverse) to improve their competitiveness in the virtual world market. The findings suggest that traditional theories of competitive advantage, which often emphasize scale or resource endowments, should be extended to include the concept of agility in technology adoption, especially for smaller, more agile firms operating in a rapidly evolving technology environment. This provides a novel theoretical perspective on how SMEs can leverage emerging technologies to improve their competitive position in the global marketplace.

6.2. *Implications for managerial practice*

The following aspects of this study make significant contributions to practice: First, the study confirms that institutional isomorphism impacts high-tech SMEs' IAMT. These results indicate that high-tech SMEs seek legitimacy, react to stakeholder pressures, and value relationships. Furthermore, SMEs can follow market leaders' best practices to gain social and technical legitimacy. This highlights the need for high-tech SMEs to be cognizant of modern marketing trends and activities of other global industry players. These findings have specific value for

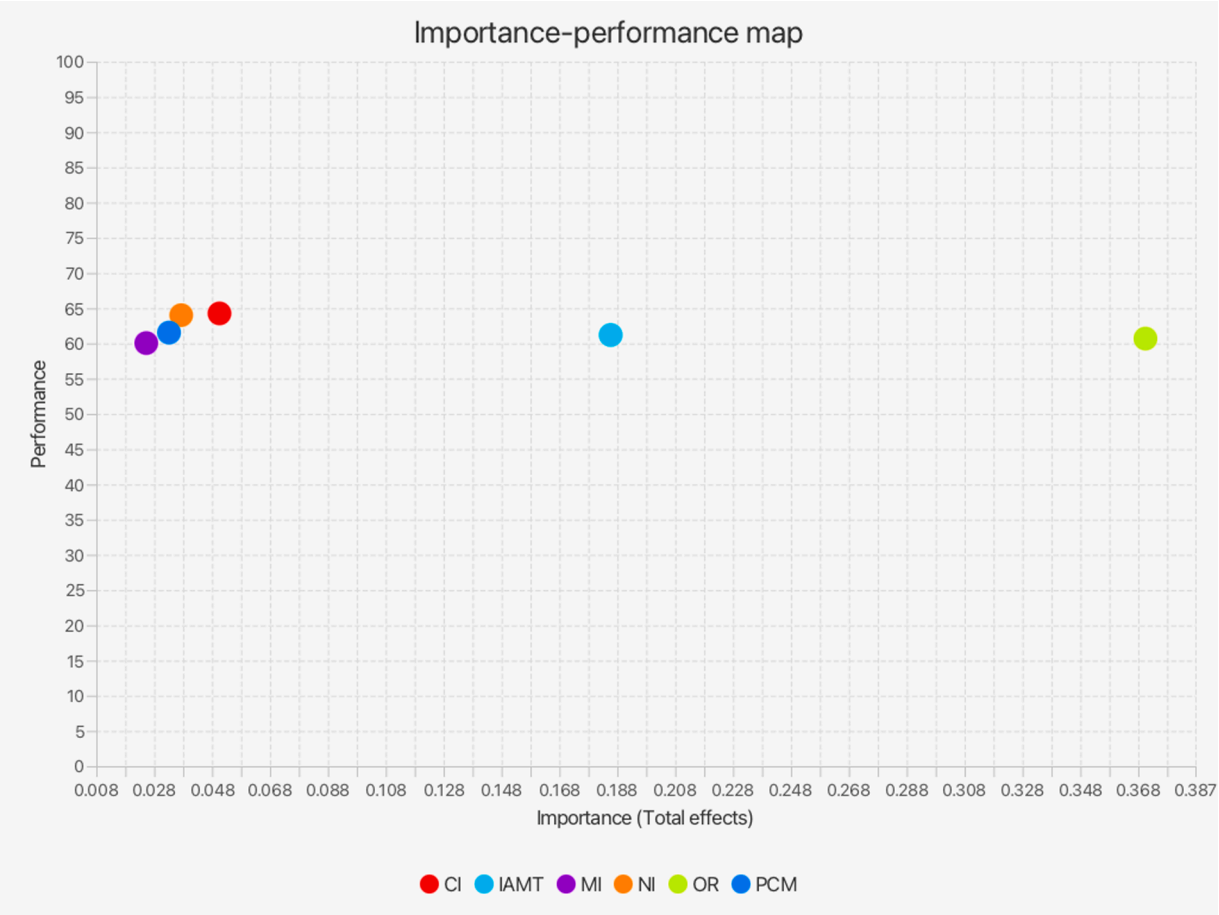


Fig. 5. Importance-performance map analysis for AMT.

policymakers and managers; specifically, the role of institutions must be strengthened to encourage more high-tech SMEs to adopt Metaverse-related technologies.

Second, the impression that PCM has been achieved for potential adopters of the core technology's utility may enhance others' ideas linked to Metaverse-related technologies. Seemingly, PCM positively impacts and thus enhances adoption intention. Therefore, we emphasize that firms can increase PCM by focusing on strategies that increase the new technology's visibility and perceived value. Additionally, firms can build communities and provide social proof of Metaverse-related technologies by encouraging users to share their experiences and success stories. This knowledge can also help managers promote IAMT and serve as a basis for future precise investigations of PCM and its involvement in AMT decisions.

Third, this study suggests that firms with higher AbC can achieve higher IMA and FP. Businesses that implement robust data analytics platforms can track users' real-time engagement in the Metaverse, enabling them to quickly adapt and customize their offerings to meet users' preferences, as evidenced by the success of companies like Roblox. Such practices also enable high-tech SMEs to consider the advantages of new technology, increase their international market visibility, and achieve their performance goals. This provides a bird's-eye view of high-tech SME management to improve AbC, which may accelerate the practical impact of decisions, particularly in cases wherein novel technologies, such as Metaverse, need to be penetrated.

Fourth, this study demonstrates that IMA might be a critical factor in SMEs' global success, but that they may need to overcome several constraints to leverage this advantage. This study's findings suggest to managers that SMEs can improve their IMA and overall performance in the global market by cultivating partnerships and collaborations and investing in training and development programs.

Fifth, this study provides practitioners with valuable insights by pinpointing the aspects contributing to AbC in high-tech SMEs. A firm's AbC (knowledge acquisition, assimilation, transformation, and exploitation) can be further strengthened to prepare it for international marketing operations. This study's outcomes can be used to design strategies and interventions to improve high-tech SMEs' ability to respond to changes in the international market.

Sixth, the results demonstrate the positive impact of AMT on IMA and FP in high-tech SMEs. In fact, these results aid managers' decision making pertaining to AMT, better respond to changes in the global market, and achieve better performance. High-tech SMEs can leverage these findings to justify and demonstrate the potential benefits of Metaverse technology. For example, a software firm specializing in gaming software may develop a Metaverse-based gaming environment wherein gamers worldwide can engage in friendly competition and explore the software's capabilities. This approach gives customers a more comprehensive understanding of the software and allows them to make informed decisions. Managers may use this setting to introduce their newest offerings to consumers, engage with them in real time, and learn more about their preferences and habits.

Finally, based on the results, we suggest that SMEs adopt various Metaverse-related technologies to transform their business models and gain a competitive edge. For example, firms are likely to use VR, AR, and spatial computing technology to create immersive product displays and virtual collaboration spaces. They also can create 3D virtual showrooms where customers can interact with goods in a realistic environment around the globe. Similarly, integrating digital twins, AI, and ML through platforms such as TensorFlow and OpenAI GPT allows for the creation of intelligent virtual assistants and personalized marketing

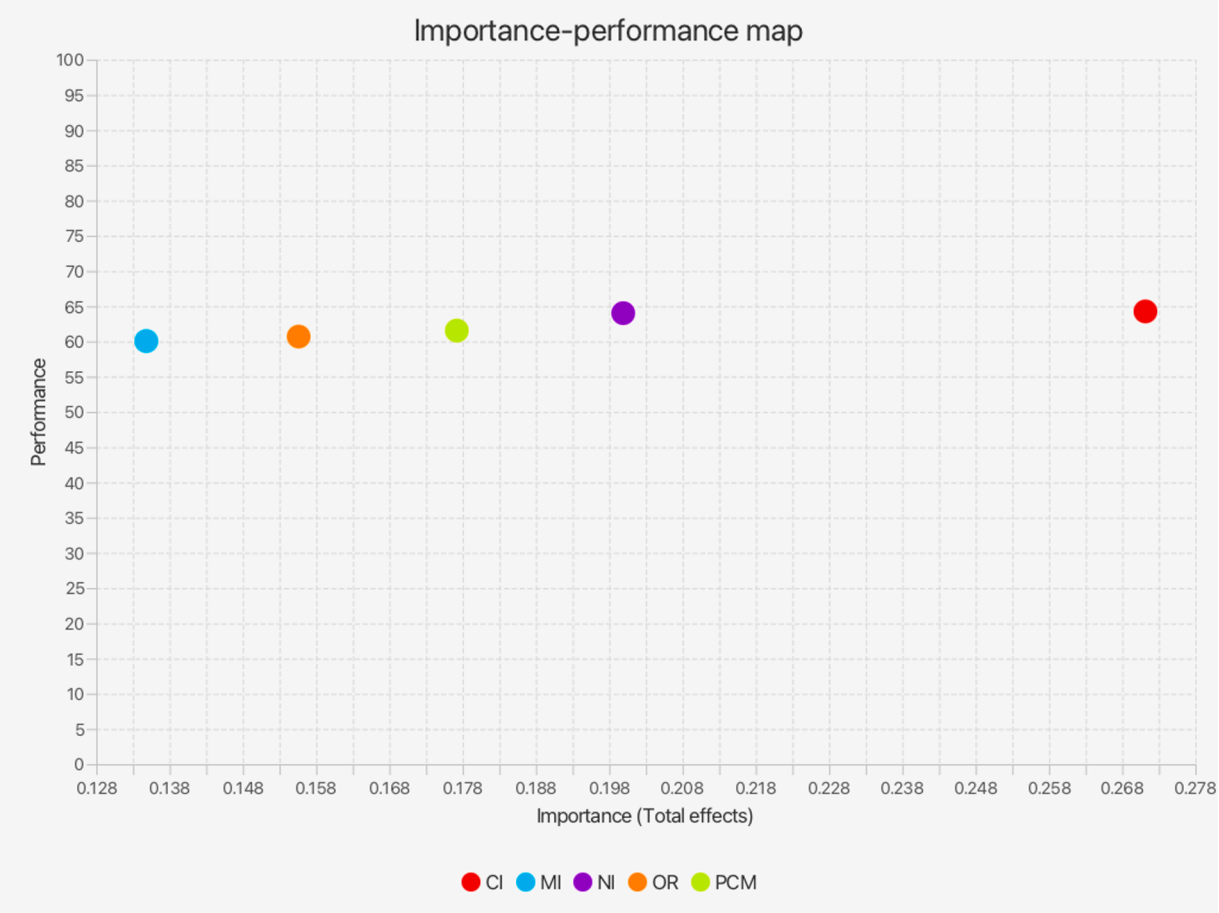


Fig. 6. Importance-performance map analysis for IAMT.

Table 13
Multi-group analysis (Age).

Hypotheses	Beta (20–30)	Beta (31–40)	Beta (above 41)	Group A		Group B		Group C	
				Diff. 20–30 to 31–40 years	p value	Diff. 20–30 to above 41 years	p value	Diff. 31–40 to above 41 years	p value
H1a= MI → IAMT	0.215	0.117	0.111	0.097	0.561	0.104	0.514	0.007	0.973
H1b= CI → IAMT	0.172	0.104	0.448	0.068	0.792	−0.276	0.189	−0.343	0.293
H1c= NI → IAMT	0.193	0.304	0.215	−0.112	0.574	−0.022	0.930	0.089	0.735
H2a= OR → IAMT	0.010	0.392	0.035	−0.382	0.061	−0.024	0.898	0.357	0.107
H2b= OR → AMT	0.430	0.051	0.427	0.379	0.097	0.003	0.992	−0.375	0.191
H3= PCM → IAMT	0.321	−0.006	0.202	0.327	0.076	0.119	0.529	−0.208	0.192
H4= IAMT → AMT	0.094	0.289	0.251	−0.195	0.312	−0.157	0.428	0.038	0.882
H5a= AMT → AbC	0.473	0.400	0.556	0.072	0.679	−0.084	0.620	−0.156	0.351
H5b= AMT → IMA	0.333	0.138	0.089	0.195	0.145	0.244	0.177	0.050	0.786
H5c= AMT → FP	0.142	0.238	0.047	−0.097	0.525	0.094	0.552	0.191	0.104
H6a= AbC → IMA	0.460	0.610	0.734	−0.15	0.441	−0.274	0.155	−0.124	0.472
H7a= AbC → FP	0.333	0.263	0.576	0.070	0.669	−0.243	0.115	−0.313	0.087
H8= IMA → FP	0.382	0.476	0.311	−0.095	0.502	0.071	0.624	0.165	0.333

campaigns. Through the strategic incorporation of such technologies into business models, managers can leverage the Metaverse's revolutionary potential to drive growth, improve customer relationships and market agility, and take a leadership role in a digitally connected world.

6.3. Limitations and future research directions

Although this study was extensive, several limitations should be addressed, as they provide a valuable foundation for future research. First, using a longitudinal design, data were collected in two waves from

high-tech SMEs in Pakistan. The results can be applied to Pakistani SMEs and other developing countries with similar industrial environments, such as its neighboring countries, and are generalizable to other technologies across various industries. Future research should use different research methods in the context of other developing countries to obtain broader results on adopting the Metaverse in other sectors, such as low-tech SMEs and service industries.

Second, future studies can investigate other variables to understand the dependent variables and improve this research model's predictive ability. OR can be further conceptualized as financial and technical readiness, which this study did not cover. The direct and indirect relationships between these aspects can be explored to measure their impact on AMT in SMEs.

Third, adoption variables are notably evolving, and new factors are expected to arise as technology advances. Therefore, it is essential to consider that these factors do not apply only to Metaverse adoption, and that each technology has a different set of adoption factors. Therefore, other factors driving adoption of the latest Metaverse-related technologies must be observed and measured.

Fourth, economic, cultural, and political differences between industrialized and developing countries significantly impact technology adoption. These factors can significantly affect AMT across industries. This study did not include these specific factors, which should be investigated in the future.

Fifth, this study assumes that the same factors influence high-tech SMEs to adopt different Metaverse-related technologies, based on different commonalities and enterprise adoption strategies. The study did not consider the individual opportunities and difficulties associated with the adoption of each technology, particularly by different businesses. Future researchers could address this issue by conducting a detailed investigation of how SMEs and other businesses are adopting specific Metaverse-related technologies. Comparing the unique advantages, challenges, and adoption strategies of each technology can provide more in-depth insights.

Finally, a high AbC helps improve the IMA and FP of SMEs; however, high AbC alone is insufficient for a firm's success. Researchers have suggested that AbC should be combined with other dynamic capabilities, such as strategic management, big data analytics, and financial capabilities, to achieve a firm's performance objectives.

7. Conclusion

The rise of the Metaverse has opened fascinating, new connections between the virtual and real worlds by creating new challenges and opportunities in various fields, as has been discussed recently [7,54]. Its adoption has raised some concerns for SMEs, however. This study provides valuable insights into AMT by high-tech SMEs in developing countries. This study is significant because high-tech SMEs substantially impact economic growth, international visibility, and social stability. By understanding the factors influencing AMT, businesses can better position themselves to exploit the opportunities and benefits of the Metaverse in the future. Furthermore, this study provides valuable insights for policymakers, managers, industry leaders, and other stakeholders interested in promoting AMT in various sectors. By understanding the factors influencing AMT, these stakeholders can develop policies and plans that better support the rapid development of high-tech SMEs. This study provides valuable insights for researchers interested in studying AMT from different perspectives.

CRediT authorship contribution statement

Fakhar Shahzad: Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Qingyu Zhang:** Conceptualization, Writing – original draft, Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

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Appendix

Questionnaire

Mimetic Isomorphism [94]

MI1: "The proportion of my business competitors that use a similar technology is with regards to the adopted technology."
MI2: "My business competitors who have entered the virtual world benefit greatly."
MI3: "Others in the same industry favorably perceive my business competitors that have adopted metaverse technological systems."
MI4: "My business competitors that have adopted metaverse-related technological systems are favorably perceived by their suppliers."
MI5: "My business competitors that have adopted metaverse-related technological systems are favorably perceived by their customers."

Coercive Isomorphism [94]

CI1: "Institutional pressures external to the firm (e.g., industry associations, customers) require the use of Metaverse-related technologies."
CI2: "My business stakeholders require the use of Metaverse-related technologies."
CI3: "The competitive conditions require our firm to use Metaverse-related technologies."

Normative Isomorphism [94]

NI1: "My business suppliers are also adopting Metaverse-related technologies."
NI2: "My business customers are also adopting Metaverse-related technologies."
NI3: "The government promotes the use of Metaverse-related technologies for small businesses."

Organizational Readiness [94]

OR1: "My firm has the financial resources for Metaverse-related technologies adoption."
OR2: "My firm has expert human resources in dealing with Metaverse-related technologies for business."
OR3: "My firm has the technical resources to use Metaverse-related technologies."
OR4: "I intend to invest and participate in Metaverse-related technologies."

Perceived critical mass [32]

PCM1: "Many of the people I work with are familiar with using Metaverse-related technologies."
PCM2: "Of the people, I communicate with regularly, many use Metaverse-related technologies."
PCM3: "A large percentage of the people I work with are willing to use Metaverse-related technologies for business."

Intention to Adopt Metaverse-related technologies [171]

IAMT1: "I intend to use Metaverse-related technologies in my business."

IAMT2: "I intend to start using Metaverse-related technologies regularly in the future in my business."
IAMT3: "I would highly recommend Metaverse-related technologies for other stakeholders to adopt for mutual benefits."

Adoption of Metaverse-related technologies [172]

AMT1: "I often use Metaverse-related technologies to manage my job-related activities."
AMT2: "I often use Metaverse-related technologies to make strategies regarding business activities."
AMT3: "I often use Metaverse-related technologies to manage, secure, and analyze digital content or business data."

Absorptive Capacity *(Higher-order Construct)* [23,173]

Acquisition

To what extent do you agree that your firm uses external resources to obtain information:
AC1: "The search for relevant information concerning our industry is an everyday business in my firm."
AC2: "The top management motivates the employees to use information sources within my firm."
AC3: "The top management expects that the employees deal with information beyond the firm."

Assimilation

To what extent do you agree that the following statements fit the communication structure in your firm:
AS1: "In my firm, ideas, technological developments, and concepts are communicated cross-departmental."
AS2: "Our management emphasizes cross-departmental support to solve problems."
AS3: "In my firm, there is a quick information flow, e.g., if a business unit obtains important information, it communicates this information promptly to all other business units or departments."
AS4: "We demand periodical cross-departmental meetings to interchange new technological developments, problems, and achievements."

Transformation

To what extent do you agree that the following statements fit the knowledge processing in your firm:
TR1: "Our employees have the ability to structure and to use collected knowledge."
TR2: "Our employees are used to absorbing new knowledge as well as preparing it for further purposes and to make it available."
TR3: "Our employees successfully link existing knowledge with new insights."
TR4: "Our employees are able to apply new knowledge in their practical work."

Exploitation

To what extent do you agree that the following statements fit the commercial exploitation of new knowledge in your firm:
EX1: "Our management supports the development of prototypes."
EX2: "My firm regularly reconsiders technologies and adapts them accordant to new knowledge."
EX3: "My firm has the ability to work more effectively by adopting new technologies."

International Marketing Agility [36]

Please rate the extent to which your business unit could respond rapidly to a new market and competitive conditions in its major foreign market compared to your competitors. (1 hardly at all and 7 very easily):
IMA1: "Entry of new competitors."
IMA2: "Emerge an innovative technology."
IMA3: "Detection of new business threats."
IMA4: "Detection of new business opportunities."

Firm Performance [23,173]

To what extent do you agree that the use of Metaverse-related technologies will lead to firm performance:
FP1: "Growth in sales."
FP2: "Return on investment."
FP3: "Operating profit margin."
FP4: "Return on equity."
FP5: "Customer retention."

Table 2
Literature summary.

References	Title	Purpose & Method	Key Findings & Implications
Abumalloh et al., [64]	The adoption of Metaverse in the retail industry and its impact on sustainable competitive advantage: moderating impact of sustainability commitment	This research paper aims to examine the factors affecting the adoption of a Metaverse in the retail sector, specifically the examination of perceived security, privacy concerns, perceived complexity, spiritual computing, infrastructure, sustainability awareness, and their impact on attitude towards Metaverse. This study uses the SEM method to identify the strength of the relationship among these variables.	The findings of this study identify the positive and negative impact of factors towards adopting Metaverse. In this study, the findings of perceived security, privacy concerns, and perceived complexity have a negative impact on the adoption of Metaverse. Also, this study found no relationship between perceived cost and decentralized technology and attitude towards Metaverse. The research also revealed a positive relationship between attitudes towards intention and intention to positively enhance product innovation, which can lead to sustainable competitive advantage. The study findings enhance the understanding of marketers and policymakers, who can design strategies based on favorable factors to adopt the Metaverse in their retail industry, enhancing their competitive advantage and sustaining their position as market leaders.

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Table 2 (*continued*)

References	Title	Purpose & Method	Key Findings & Implications
Adhini & Prasad, [69]	Perceptions and drivers of the Metaverse adoption: A mixed-methods study	This study aims to analyze and identify behavioral factors that can enhance the intention to use Metaverse. It aims to understand different dimensions influencing the adoption of MV, focusing on the presence of telepresence and social presence facilitating conditions and regulatory support. For this purpose, the study used a mixed-method approach.	The study's key findings include the importance of telepresence and social presence in adopting MV and variations in attitudes and intentions across age, gender, and income level. Organizations should work to make it more user-friendly and enjoyable, cultivate user trust, and emphasize virtual presence, which can provide a roadmap for developing Metaverse services.
Agnihotri et al., [68]	Driving Metaverse adoption in the hospitality industry: an upper echelon perspective	This study uses PLS-SEM to analyze how CEOs in the hospitality industry's past and future temporal focus influences their intention to invest in Metaverse technology and to identify a mechanism under the boundary conditions of perceived competitiveness.	The findings of this study suggest that a CEO's past temporal focus decreases a firm's technology orientation, whereas futural temporal focus increases it. Additionally, perceived competitive pressure moderates this relationship by reducing the negative impact of past temporal focus on technology orientation and enhancing the future temporal focus. Through adopting the Metaverse, this study has significantly contributed to the success of the hospitality industry.
Arpaci & Bahari, [62]	Investigating the role of psychological needs in predicting the educational sustainability of Metaverse using a deep learning-based hybrid SEM-ANN technique	This study investigated key factors of psychological needs predicting educational sustainability in Metaverse. This study employed a hybrid method integrating CB-SEM and deep ANN model to meet the objectives.	The results show that psychological needs predict the educational sustainability of Metaverse. The need for autonomy and hedonic motivation play important roles in predicting educational sustainability in Metaverse. Developers of Metaverse environments can benefit from understanding the importance of hedonic motivation and autonomy in predicting educational sustainability.
Arpaci et al., [61]	Understanding the social sustainability of the Metaverse by integrating UTAUT2 and big five personality traits: A hybrid SEM-ANN approach	This study's purpose was to identify the factors that can contribute to and enhance companies', developers', designers', and other institutions' adoption of Metaverse. For this purpose, the study used motivational factors, personality traits, and social sustainability, using the UTAUT2 theory and five personality traits to explain the social sustainability of Metaverse.	The key findings of this study are that companies should provide training and enhance the friendliness of the Metaverse to achieve its sustainable use. By doing so, companies can enhance the overall long-term goal selection and encourage employees to work with more motivation to improve and adopt Metaverse to achieve these goals. Also, people should adopt such technologies daily to make their social impact significant.
Chen et al., [27]	The metaverse in supply chain knowledge sharing and resilience contexts: An empirical investigation of factors affecting adoption and acceptance	This study aims to develop a conceptual framework based on the unified theory of acceptance and use of technology (UTAUT), network theory views, and Metaverse's characteristics among the Chinese firms' supply chain networks. Based on the survey data, this study applied covariance-based structural equation modeling.	The results show that several factors, such as performance expectations, facilitating conditions, and establishing initial trust among supply chain partners, can drive their intention to adopt knowledge sharing to improve supply chain resilience. Similarly, sensory feedback was found to be a core feature of Metaverse adoption intentions in the supply chain. This finding could give supply chain managers a greater perspective on developing strategies for implementing Metaverse technologies for better visibility.
Fazio et al., [67]	Metaverse and tourism development: issues and opportunities in stakeholders' perception	The main purpose of this study is to identify the important role of Metaverse by different stakeholders in the tourism sector, including private and public actors, skill providers, and managers. This study used Q-methodology to understand how these perceptions impact promotion strategy for tourist destinations.	The findings reveal various perceptions among different stakeholder groups regarding the Metaverse. According to this study, the tourism industry is in the early stages of adopting the Metaverse. This study identifies that stakeholders evaluate the overall benefits and drawbacks when adoption increases. The stakeholder decision-making process is important in planning for sustainable destinations, good governance, and effective destination management.
Gil-Cordero et al., [26]	Do small- and medium-sized companies intend to use the Metaverse as part of their strategy? A behavioral intention analysis	This article analyzes the factors influencing small—and medium-sized industries' decision to adopt Metaverse. This study explicitly identifies effort expectancy and performance expectancy concerning business satisfaction. The study used the SEM method to analyze the relationship between constructs.	The key findings of this study indicate that the valuation of the effort to be made and expected performance by the company does not directly determine their intention to use immersive technology in their strategic behavior. The findings show that obtaining information, reducing uncertainty, and analyzing the competition can enhance satisfaction. This study contributes not only to the adoption of Metaverse in SMEs but also to its broader acceptance and utilization across different domains, providing a way to adopt different technological advancements and societal benefits.
Gupta et al., [28]	Are we ready for Metaverse adoption in the service industry? Theoretically exploring the barriers to successful adoption	Identification and analysis of the barriers preventing the adoption of Metaverse in service firms. Six groups of the different obstacles were analyzed by using Force filed theory, and the TOE framework was used	This research identifies 36 barriers from these 6 groups. This study found that security and privacy are among the most influential barriers to adopting Metaverse in the service industry. The study provides a valuable roadmap for managers to

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Table 2 (*continued*)

References	Title	Purpose & Method	Key Findings & Implications
		to understand the uses and research to identify the causal relationship by using FAHP and FDEMATEL	identify these barriers and make strategies to overcome them for a better and more positive effect.
Hasani et al., [19]	Towards a framework for successful metaverse adoption in Small and Medium-sized Enterprises: An exploratory study	This exploratory study used the technology-organization-environment (TOE) framework to explore the factors influencing Metaverse adoption intentions and their impact on SME performance. Cross-sectional data was used, and a regression model was applied to test the model.	The results reveal several significant factors influencing SMEs' intentions to adopt Metaverse. Notably, anxiety, security, and privacy did not affect adoption intentions. On the contrary, the intention to adopt Metaverse positively correlates with SME performance. The paper presents insights for policymakers on the factors influencing the priority decision-making of Metaverse adoption by Canadian SMEs.
Kumar et al., [12]	With enterprise Metaverse comes great possibilities! Understanding Metaverse usage intention from an employee perspective	This study explored the dynamics of organizational adoption of the enterprise Metaverse, focusing on its integration with existing systems, workplace culture, and employee well-being. For this purpose, it used diffusion of innovation theory to identify its relationship with enterprise Metaverse usage intention.	The study's findings support the idea that the dimension of diffusion innovation theory is significantly associated with the usage of Metaverse enterprise. This study also finds the mediating role of digital agility between these factors of adoption intention and DIT theory dimensions. This study's findings guide the organization, connecting researchers and organizations to facilitate a responsible and impactful adoption of Metaverse in an enterprise, which will benefit the organization's growth.
Kumar et al., [13]	Risking it all in the Metaverse ecosystem: forecasting resistance towards the enterprise Metaverse	This study investigates organizations' non-adoption intentions regarding the enterprise Metaverse and identifies risk factors.	The findings of this study identify that performance, security, and psychological risk are positively associated with intention not to use Metaverse. Also, the study identifies the risk related to investments leading to the non-adoption of MV. Where social infrastructure and technological dependence are not significant to the adoption of MV in enterprise From a B2B perspective, this study will provide a roadmap to remove these barriers to adopting the Metaverse.
Kumari et al., [14]	A text mining approach to explore factors influencing consumer intention to use Metaverse platform services: Insights from online customer reviews	This study explores factors impacting consumer intention to use mobile-based Metaverse platform services. To this end, it uses an LDA-based topic modeling approach to analyze online user reviews of the Highrise Virtual Metaverse mobile platform.	Study findings show that visually appealing virtual environments and enjoyment from virtual events positively enhance consumer intention to use Metaverse platforms on mobile. These services also enhance social interaction. During the study analysis, joy was the users' most frequently expressed emotion. Some factors, like mobile app issues, user account problems, and the watch ad-to-earn feature, negatively impact intention in this study.
Lu et al., [31]	How to trigger user's willingness to participate in the Metaverse? An exploration of the significant factors of the Metaverse	This research investigates the user's willingness to participate in the Metaverse through the lenses of self-efficacy and TRA.	The findings reveal that respondents' willingness to participate in the Metaverse is significantly influenced by their presence in Second Life and play-to-earn activities. Further, 3D interactivity increases intentions to participate by affecting their presence in Second Life and engagement in play-to-earn activities. This study's findings will enhance user presence, improve interactivity, increase engagement, and integrate decentralized mechanisms. They will also provide a way to invest in such technologies.
Mirza et al., [70]	Clustering potential Metaverse users with the use of a value-based framework: Exploiting perceptions and attitudes on the use and adoption of Metaverse for bold propositions	The purpose of this study is to study users of the Metaverse and obtain their views on various key factors of the adoption, such as core concepts, readiness, ease of use, intentions to adopt Metaverse, and value realization.	The findings of this study contribute to understanding current and potential Metaverse users, especially those from Saudi Arabia. Therefore, this research will help to develop Metaverse ecosystem, enhance their functionality, and provide the best experience for users. This study also contributes to the adoption of the Metaverse platform by demonstrating a value-based framework that includes Skeptical, Unaware, and Optimists.
Natarajan et al., [71]	Uses and gratifications of Metaverse: understanding the user adoption factors through a mixed method approach	This study aims to explore the uses and gratification factors affecting user adoption and Metaverse's recommendation from the management perspective. It uses a mixed-method approach to validate these factors.	The study identifies the variables that significantly impact the intention to adopt and recommend Metaverse: escapism, social interaction, convenient navigability, and telepresence. It provides a roadmap for enhancing this intention.
Park [30]	Examining Metaverse game platform adoption: Insights from innovation, behavior, and coolness	This study aimed to identify factors influencing players' acceptance of Metaverse game platforms using research on four user-oriented approaches: 1. TPB, 2. Hedonic and Utilitarian Concept, 3. Diffusion of innovation theory, and 4. Coolness concept. The proposed model was tested using structural equation modeling and confirmatory factor analysis.	The research findings explore key factors affecting player intention to use Metaverse game platforms, including attitude, enjoyment, compatibility, tribality, observability, and coolness factors such as attractiveness and subcultural appeal. This identification gives developers a roadmap to create immersive game experiences, facilitate social

(*continued on next page*)

Table 2 (*continued*)

References	Title	Purpose & Method	Key Findings & Implications
			interaction, and enhance the features that can increase players' enjoyment levels. Strategies to make it user-friendly and remove complexity.
Shukla et al., [29]	The future of Metaverse adoption: A behavioral reasoning perspective with a text-mining approach	This study aims to identify both supportive and opposing factors for adopting the Metaverse; for that purpose, this research paper used STM, which is Structural Topic Modelling, to analyze different research articles with a sentiment analysis to understand the sentiments towards MV.	These research findings explain in detail that identifying flexibility, brand experience, human centricity, and virtual retail experience are the enablers of acceptance of the Metaverse, while regulatory issues, like usage barriers, financial investment, and skepticism, act as barriers to its adoption. The findings of this study contribute to the existing knowledge for marketers who want to adopt the Metaverse in their operations.
Wu & Yu, [63]	Investigating Users' Acceptance of the Metaverse with an Extended Technology Acceptance Model	This study aims to extend the technology acceptance model by integrating social and psychological constructs such as social interaction, social presence, conformity, emotional attachment, flow, and perceived enjoyment. The data for this study were gathered through an online survey, and a partial least squares structural equation model was used to test the hypotheses.	The findings proved that emotional attachment, perceived enjoyment, and social presence positively influence the perceived ease of use of Metaverse technology. Moreover, affective attachment, enjoyment, and ease of use positively affect perceived usefulness. Similarly, attitude, perceived usefulness, conformity, and social interaction positively influence the intention to use the Metaverse. Contrary to previous research, the results of this study show that social presence has no significant effect on perceived usefulness. These findings have important implications for TAM models' theoretical and practical extension in this context.

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